

Signed Angle Rigid Graphs for Network Localization and Formation Control

Jinpeng Huang and Gangshan Jing

Abstract—Graph rigidity theory studies the capability of a graph embedded in the Euclidean space to constrain its global geometric shape via local constraints among nodes and edges, and has been widely exploited in network localization and formation control. In recent years, the traditional rigidity theory has been extended by considering new types of local constraints such as bearing, angle, ratio of distance, etc. Among them, the signed angle constraint has received extensive attention, since it is practically measurable and independent of the global coordinate frame. However, the relevant studies always consider special graph structures, which are sufficient but not necessary for signed angle rigidity. This paper presents a comprehensive combinatorial analysis in terms of graphs and angle index sets for signed angle rigidity. We show that Laman graphs equivalently characterize minimally signed angle rigid graphs. Moreover, we propose a method to construct the minimal set of signed angle constraints in a Laman graph to effectively ensure signed angle rigidity. These results are finally applied to distributed network localization and formation stabilization problems, respectively, where each agent only has access to signed angle measurements.

Index Terms—Graph rigidity theory, rigid formation control, network localization

I. INTRODUCTION

The traditional graph rigidity theory studies the characterizability of a geometric shape by distance constraints among neighboring vertices in the graph, and has been widely employed in multi-agent coordination problems such as network localization [1–3] and formation control [4–8]. Inspired by this, rigidity theories based on other types of constraints (such as bearings [9, 10], angles [11, 12], ratio of distance [13], etc.) have been further developed to achieve network localization [14–18] and formation stabilization [9–13, 19, 20] via different sensing measurements. A geometric shape specified by different types of constraints may require different graphical conditions to ensure a unique shape [21]. As a result, different conditions on the network topology have been proposed depending on the type of measurements utilized.

In distance-based multi-agent formation control, infinitesimal distance rigidity has been adopted most widely since it has

an equivalent rank condition on the rigidity matrix [22], which plays a vital role in the gradient-based controller analysis [5, 23–25]. On the other hand, although global distance rigidity implies a unique shape, and has been favored in network localization [1], it usually requires a large number of edges. Moreover, the formation controller with global convergence under globally rigid graphs is yet to be designed [26].

In [10], infinitesimal bearing rigidity was shown to share the same graphical condition¹ as infinitesimal distance rigidity (i.e., Laman graphs [27, 28]), while guaranteeing that the geometric shape is uniquely determined by bearings. Taking this advantage, the authors in [10, 16] proposed novel distributed protocols for bearing-based formation stabilization and sensor network localization, where almost global convergence is achieved. These works modeled agents as point particles, a formation is treated as a framework in the Euclidean space and bearings are expressed in the global coordinate frame. A more practical formulation of bearing rigidity theory treats agents as rigid bodies and considers frameworks embedded in the Special Euclidean group [29], where bearings are expressed in local coordinate frames. However, the graphical conditions for this type of bearing rigidity have yet to be established.

Recently, angle rigidity theory has garnered significant interest, as angle measurements are independent of the coordinate frame [14, 17, 18, 30–32], and an angle-constrained formation exhibits higher degrees of freedom than distance and bearing-based approaches [11, 12]. In [11], the authors established the equivalence between infinitesimal angle rigidity, infinitesimal bearing rigidity, and infinitesimal distance rigidity in the plane. However, it is unknown whether the shape of an infinitesimally angle rigid framework can be uniquely determined by angles. In [12], the authors reformulated angle rigidity theory by considering signed angle constraints. However, the signed angle rigidity property was studied on angularities², which are essentially frameworks with specified signed angles. Such a setting benefits for excluding redundant angles, but in the meanwhile, hinders the connection between the graph structure and the signed angle rigidity property.

In this paper, we study the conditions on graphs and the minimal set of signed angle constraints required to ensure signed

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¹Throughout this paper, the “graphical condition” for some rigidity property refers to the condition on the graph $\mathcal{G}$ for the framework $(\mathcal{G}, p)$ to have this rigidity property.

²Angularities are defined by $\mathbb{A}(\mathcal{V}, \mathcal{A}, p)$ where $\mathcal{V}$ is the vertex set, $\mathcal{A} \subset \mathcal{V} \times \mathcal{V} \times \mathcal{V}$ is the angle index set, p is the vertex configuration, and the edge exists between i and j if they are incorporated in a common angle constraint.

angle rigidity and uniqueness of the shape. Similar issues related to angle rigidity theory have been investigated in the literature. In [33, 34], the authors explored the combinatorial conditions of rigidity under angle constraints. In [11, 12, 18], the authors reported a class of particular graphs which ensure a unique shape under angle constraints. Nevertheless, these works fail to characterize the equivalent graphical condition for angle rigidity and the minimal set of angle constraints to guarantee a unique shape. Considering the large body of existing works in bearing rigidity theory, we believe that establishing the relationship between signed angle rigidity and bearing rigidity would be a worthwhile exploration.

Our contributions can be summarized as follows. (i). Enlightened by angle rigidity theory in [11, 12], we develop the signed angle rigidity theory in the context of frameworks. Under this setting, (infinitesimal, global) signed angle rigidity and (infinitesimal, global) bearing rigidity are completely equivalent, implying that the underlying graph for a signed angle rigid framework always contains a Laman subgraph; see Section II–III. (ii). Similarly to the bearing case, we prove that infinitesimal signed angle rigidity is sufficient and necessary for a framework to determine its shape uniquely; see Theorem 3. (iii). We provide an analysis on the set of angle constraints and develop Algorithm 1 to generate a minimal angle index set to characterize the rigidity properties of a framework; see Section IV. (iv). Based on the theory developed, we solve the problems of network localization and formation stabilization, respectively, where each agent only has access to local signed angle measurements associated with its neighbors. Two distributed algorithms with global convergence guarantee are proposed, respectively, where the conditions on sensing graphs contain those in [11, 12, 17, 18]; see Section V–VI.

This paper is structured as follows. Section II reviews bearing rigidity theory and introduces the concept of signed angle rigidity theory for frameworks in the plane. Section III establishes the equivalence on bearing rigidity and signed angle rigidity. Section IV provides a combinatorial analysis for minimal angle index sets. Section V–VI demonstrate the application of our results to network localization and formation stabilization. Section VII summarizes the entire paper.

Notation: In this paper, let the set of real numbers be $\mathbb{R}$, d -dimensional Euclidean space be $\mathbb{R}^d$, $\|\cdot\|$ be the Euclidean norm, $|A|$ be the number of the elements in the set A , and $X^\top$ be the transpose of the matrix X . Denote $\otimes$ by the Kronecker product. $\text{Null}(X)$ and $\text{rank}(X)$ are defined as the null space and rank of the matrix X , respectively. $\mathbf{1}_n \triangleq [1, \dots, 1]^\top$. Define $I_n \in \mathbb{R}^{n \times n}$ as the identity matrix, $A \setminus B$ as the set that consists of the elements belonging to A but not to B , and $\mathcal{R}_o(\theta)$ as the 2-dimensional rotation matrix associated with $\theta \in [0, 2\pi)$. For a vector $x \in \mathbb{R}^2$, $x^\perp \triangleq \mathcal{R}_o(\frac{\pi}{2})x$. For $X_i \in \mathbb{R}^{a \times b}$, $i = 1, \dots, q$, $\text{diag}(X_i) \triangleq \text{blockdiag}\{X_1, \dots, X_q\} \in \mathbb{R}^{qa \times qb}$.

We consider the undirected graph $\mathcal{G} \triangleq (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} \triangleq \{1, \dots, n\}$ is the vertex set satisfying $n \geq 3$, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$

TABLE I
NOTATION

Notation	Description
$\bar{\mathcal{T}}_{\mathcal{G}}$	Angle index set satisfying $\bar{\mathcal{T}}_{\mathcal{G}} \subseteq \mathcal{T}_{\mathcal{G}}$
$\mathcal{T}_{\mathcal{G}}$	The set indexing the whole angles in the graph $\mathcal{G}$
$\mathcal{H}(i, j, \mathcal{V})$	Hash function to encode the edge (i, j) of $\mathcal{G}$
$B_{\mathcal{G}}(\cdot)$	Bearing rigidity function under the graph $\mathcal{G}$
$S_{\mathcal{G}}(\cdot)$	Signed angle rigidity function under the graph $\mathcal{G}$
$R_B(p)$	Bearing rigidity matrix under the framework $(\mathcal{G}, p)$
$\mathcal{E}(p)$	Configuration set that forms the same shape as p
$\mathcal{E}_{\mathcal{G}}(p)$	Configuration set of frameworks with the same signed angles as $(\mathcal{G}, p)$
$\mathcal{E}_{\bar{\mathcal{T}}_{\mathcal{G}}}(p)$	Configuration set of frameworks with the same signed angles indexed by $\bar{\mathcal{T}}_{\mathcal{G}}$ as $(\mathcal{G}, p)$
$R_S^{\bar{\mathcal{T}}_{\mathcal{G}}}(p)$	Signed angle rigidity matrix under the framework $(\mathcal{G}, p)$ and the angle index set $\bar{\mathcal{T}}_{\mathcal{G}}$
$\mathcal{N}_{\mathcal{G}}^i$	Neighbor set of node i in $\mathcal{G}$. If $\mathcal{G}$ is clear from the context, it is simplified by $\mathcal{N}_i$
$r^{-1}(r(p))$	The preimage of $r(p)$ under the smooth function $r(\cdot)$ and the configuration $p \in \mathbb{R}^{dn}$
$\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{G}})$	Angle index graph $\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{G}}) = (\mathcal{V}_A(\mathcal{G}), \mathcal{E}_A(\bar{\mathcal{T}}_{\mathcal{G}}))$ under the set $\bar{\mathcal{T}}_{\mathcal{G}}$ and $\mathcal{G}$. If $\mathcal{G}$ and $\bar{\mathcal{T}}_{\mathcal{G}}$ are clear from the context, it is simplified by $\mathcal{G}_A = (\mathcal{V}_A, \mathcal{E}_A)$

is the edge set, and $(i, j) \in \mathcal{E}$ is equivalent to $(j, i) \in \mathcal{E}$ in $\mathcal{G}$. The set of neighbors of the vertex j in $\mathcal{G}$ is denoted by $\mathcal{N}_{\mathcal{G}}^j \triangleq \{i \in \mathcal{V} : (j, i) \in \mathcal{E}\}$. If $\mathcal{G}$ is clear from the context, we simply denote $\mathcal{N}_{\mathcal{G}}^j$ by $\mathcal{N}_j$. The undirected complete graph with the same number of nodes as $\mathcal{G}$ is denoted by $\mathcal{K}$. A graph $\bar{\mathcal{G}} \triangleq (\bar{\mathcal{V}}, \bar{\mathcal{E}})$ is said to be a subgraph of $\mathcal{G}$ if $\bar{\mathcal{V}} \subseteq \mathcal{V}$ and $\bar{\mathcal{E}} \subseteq (\bar{\mathcal{V}} \times \bar{\mathcal{V}}) \cap \mathcal{E}$. A subgraph is said to be a spanning subgraph if it is connected and $\bar{\mathcal{V}} = \mathcal{V}$.

Given an undirected graph $\mathcal{G}$, an undirected path between $l_1 \in \mathcal{V}$ and $l_{n_P} \in \mathcal{V}$ is a sequence of edges of the form $(l_1, l_2), \dots, (l_{n_P-1}, l_{n_P})$. In this paper, for the sake of analysis, we use the subgraph $\mathcal{P} \triangleq (\mathcal{V}_P, \mathcal{E}_P)$ to represent the corresponding undirected path, where $\mathcal{V}_P \triangleq \{l_1, \dots, l_{n_P}\} \subseteq \mathcal{V}$ and $\mathcal{E}_P \triangleq \{(l_1, l_2), \dots, (l_{n_P-1}, l_{n_P})\} \subseteq \mathcal{E}$. An undirected cycle is an undirected path in $\mathcal{G}$ that starts and ends at the same node. We use the subgraph $\mathcal{C} \triangleq (\mathcal{V}_C, \mathcal{E}_C)$ to represent the corresponding undirected cycle, where $\mathcal{V}_C \triangleq \{l_1, \dots, l_{n_P}\} \subseteq \mathcal{V}$ and $\mathcal{E}_C \triangleq \{(l_1, l_2), \dots, (l_{n_P-1}, l_{n_P}), (l_{n_P}, l_1)\} \subseteq \mathcal{E}$. Table I shows the principal variables used throughout this paper.

II. PRELIMINARIES

In this section, we briefly review some background on bearing rigidity theory [10], define signed angle rigidity in the context of frameworks, and introduce some concepts related to the index set of angle constraints.

A. Graph Rigidity Theory

To begin with, we introduce some important notions in rigidity theory based on a smooth rigidity function, which are drawn from the results of [10, 11, 22]. Given different smooth rigidity functions, different rigidity theories can be

developed accordingly, e.g., bearing rigidity theory and signed angle rigidity theory.

A framework $(\mathcal{G}, p)$ in $\mathbb{R}^d$ is characterized by an undirected graph $\mathcal{G}$ and a configuration $p = [p_1^\top, \dots, p_n^\top]^\top \in \mathbb{R}^{dn}$, where p_i is the coordinate mapped to the vertex $i, i = 1, \dots, n$ with the assumption that $p_i \neq p_j$ for any $j \in \{1, \dots, n\} \setminus \{i\}$. If $p_1, \dots, p_n$ do not belong to a hyperplane of $\mathbb{R}^d$, we refer p to be *non-degenerate*, and the framework $(\mathcal{G}, p)$ is said to be a *non-degenerate framework*. Given a spanning subgraph $\bar{\mathcal{G}}$ of $\mathcal{G}$, $(\bar{\mathcal{G}}, p)$ is said to be the *spanning subframework* of $(\mathcal{G}, p)$.

Given a framework $(\mathcal{G}, p)$, a smooth *rigidity function* $r_{\mathcal{G}}(\cdot) : \mathbb{R}^{dn} \rightarrow \mathbb{R}^s$ is a function of $p \in \mathbb{R}^{dn}$, and each entry of $r_{\mathcal{G}}(p)$ corresponds to a constraint in $(\mathcal{G}, p)$, here s is the number of constraints. Based on $r_{\mathcal{G}}(\cdot)$, a series of concepts can be given.

Definition 1: In $\mathbb{R}^d$, a framework $(\mathcal{G}, p)$ is said to be *rigid* if there exists a neighborhood U_p of p , such that for any $q \in r_{\mathcal{G}}^{-1}(r_{\mathcal{G}}(p)) \cap U_p$, it always holds $r_{\mathcal{K}}(q) = r_{\mathcal{K}}(p)$.

Definition 2: In $\mathbb{R}^d$, a framework $(\mathcal{G}, p)$ is said to be *globally rigid* if $r_{\mathcal{G}}^{-1}(r_{\mathcal{G}}(p)) = r_{\mathcal{K}}^{-1}(r_{\mathcal{K}}(p))$.

Definition 3: In $\mathbb{R}^d$, a rigid framework $(\mathcal{G}, p)$ is said to be *minimally rigid* if there does not exist any rigid spanning subframework $(\bar{\mathcal{G}}, p)$ of $(\mathcal{G}, p)$ such that $|\bar{\mathcal{E}}| < |\mathcal{E}|$.

By definition, global rigidity implies rigidity. An instantaneously continuous motion $\dot{p} \triangleq v = [v_1^\top, \dots, v_n^\top]^\top$ is said to be *infinitesimal* if it preserves the invariance of $r_{\mathcal{G}}(p)$, i.e.,

$$\dot{r}_{\mathcal{G}}(p) = \frac{\partial r_{\mathcal{G}}(p)}{\partial p} v = 0. \quad (1)$$

A motion is said to be *trivial* if it preserves rigidity for any framework. For example, if each entry of $r_{\mathcal{G}}(p)$ is a distance, i.e., $\|p_i - p_j\|$, then the trivial motions consist of uniform rotations and translations of the whole framework [35].

Definition 4: In $\mathbb{R}^d$, a framework $(\mathcal{G}, p)$ is said to be *infinitesimally rigid* if all of its infinitesimal motions are trivial.

The *rigidity matrix* is defined as $R(p) \triangleq \frac{\partial r_{\mathcal{G}}(p)}{\partial p}$. Then, (1) can be rewritten as $\dot{r}_{\mathcal{G}}(p) = R(p)\dot{p} = 0$. It can be obtained that a framework $(\mathcal{G}, p)$ in $\mathbb{R}^d$ is infinitesimally rigid if and only if $\text{rank}(R(p)) = dn - T$, where T is the dimension of the space spanned by trivial motions.

So far, we have introduced some relevant rigidity concepts based on a given rigidity function. Throughout this paper, we mainly focus on the case in $\mathbb{R}^2$. In following subsections, we will introduce two specific rigidity functions, i.e., bearing rigidity function and signed angle rigidity function, based on which similar rigidity concepts can be induced naturally.

B. Bearing Rigidity

Given a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, according to [10], the *bearing rigidity function* is described as:

$$B_{\mathcal{G}}(p) = [\dots, b_{ji}^\top, \dots]^\top, (j, i) \in \mathcal{E}, \quad (2)$$

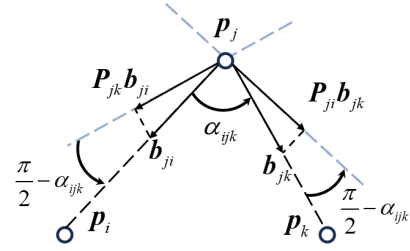


Fig. 1. Example of the bearing and signed angle among three agents, where blue dashed lines represent the orthogonal complements of b_{ji} and b_{jk} , respectively.

where $b_{ji} \triangleq b_{ji}(p) = \frac{p_i - p_j}{\|p_i - p_j\|} \in \mathbb{R}^2$ is the bearing vector from p_j to p_i (see Fig. 1). The *bearing rigidity matrix* is

$$R_B(p) \triangleq \text{diag} \left(\frac{P_{ji}}{\|e_{ji}\|} \right) \bar{H} \in \mathbb{R}^{2|\mathcal{E}| \times 2n}, \quad (3)$$

where $e_{ji} \triangleq p_i - p_j$, $P_{ji} \triangleq P(b_{ji}) = I_2 - b_{ji}b_{ji}^\top$ is an orthogonal projection matrix projecting any vector onto the orthogonal space of b_{ji} , $\bar{H} \triangleq (H \otimes I_2)$, $H = [h_{ji}]$ is the incidence matrix whose rows are indexed by edges of $\mathcal{G}$ and columns are indexed by vertices of $\mathcal{G}$ with an orientation³: $h_{ji} = 1$ if vertex i is the end of the j th edge, $h_{ji} = -1$ if vertex i is the start of the j th edge, and $h_{ji} = 0$ otherwise.

Since bearing-preserving motions of a framework in $\mathbb{R}^2$ include 2-dimensional uniform translations and 1-dimensional uniform scalings [10], a framework is infinitesimally bearing rigid if and only if $\text{rank}(R_B(p)) = 2n - 3$.

C. Signed Angle Rigidity

Given a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, according to [12], the *signed angle rigidity function* is defined as:

$$S_{\mathcal{G}}(p) = [\dots, \alpha_{ijk}, \dots]^\top, (i, j, k) \in \mathcal{T}_{\mathcal{G}}, \quad (4)$$

where $\alpha_{ijk} \triangleq \alpha_{ijk}(p) \in [0, 2\pi)$ describes the signed angle from b_{ji} to b_{jk} in the counter-clockwise direction (see Fig. 1), and is computed by

$$\alpha_{ijk} = \begin{cases} \arccos(b_{jk}^\top b_{ji}) & \text{if } b_{jk}^\top b_{ji} \geq 0, \\ 2\pi - \arccos(b_{jk}^\top b_{ji}) & \text{otherwise,} \end{cases} \quad (5)$$

$\mathcal{T}_{\mathcal{G}} = \{(i, j, k) \in \mathcal{V}^3 : (j, i), (j, k) \in \mathcal{E}, i < k\}$ is a set indexing all signed angles in $(\mathcal{G}, p)$, here (i, j, k) corresponds to the signed angle α_{ijk} . By differentiating $S_{\mathcal{G}}(p)$ w.r.t. p , one has the *signed angle rigidity matrix*:

$$R_S^{\mathcal{T}_{\mathcal{G}}}(p) \triangleq \text{diag} \left(-\frac{1}{\sin \alpha_{ijk}} \right) R_b(p) R_B(p) \in \mathbb{R}^{|\mathcal{T}_{\mathcal{G}}| \times 2n}, \quad (6)$$

³Here, the orientation assigned to the graph here is arbitrary, and does not affect the validity of the results.

where $R_B(p)$ is the bearing rigidity matrix and $R_b(p) \in \mathbb{R}^{|\mathcal{T}_G| \times 2|\mathcal{E}|}$ can be written as

$$\alpha_{ijk} \begin{bmatrix} \cdots & b_{ji} & \cdots & b_{jk} & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \mathbf{0} & b_{jk}^\top & \mathbf{0} & b_{ji}^\top & \mathbf{0} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$$

whose rows are indexed by $\mathcal{T}_G$ and columns are indexed by the edges of $\mathcal{G}$.

Note that the existence of $\sin \alpha_{ijk} = 0$ in (6) seems to render the signed angle rigidity matrix $R_S^{\mathcal{T}_G}(p)$ invalid. In fact, differentiating both sides of (5) w.r.t. time leads to

$$\dot{\alpha}_{ijk} = \frac{b_{jk}^\top P_{ji}}{-\|e_{ji}\| \sin \alpha_{ijk}} \dot{p}_i + \frac{b_{ji}^\top P_{jk}}{-\|e_{jk}\| \sin \alpha_{ijk}} \dot{p}_k - \left(\frac{b_{jk}^\top P_{ji}}{-\|e_{ji}\| \sin \alpha_{ijk}} + \frac{b_{ji}^\top P_{jk}}{-\|e_{jk}\| \sin \alpha_{ijk}} \right) \dot{p}_j. \quad (7)$$

It can be observed from Fig. 1 that the terms $b_{jk}^\top P_{ji}$ and $b_{ji}^\top P_{jk}$ in (7) can be expressed as $-b_{ji}^\top \mathcal{R}_o(\frac{\pi}{2}) \sin \alpha_{ijk}$ and $b_{jk}^\top \mathcal{R}_o(\frac{\pi}{2}) \sin \alpha_{ijk}$, respectively. Therefore, we have another expression for the signed angle rigidity matrix.

Lemma 1: The rigidity matrix (6) can be expressed as

$$R_S^{\mathcal{T}_G}(p) = \bar{R}_{\mathcal{T}_G}(p) \bar{H}, \quad (8)$$

where $\bar{R}_{\mathcal{T}_G}(p) \in \mathbb{R}^{|\mathcal{T}_G| \times 2|\mathcal{E}|}$ is

$$\alpha_{ijk} \begin{bmatrix} \cdots & b_{ji} & \cdots & b_{jk} & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \mathbf{0} & \frac{b_{ji}^\top \mathcal{R}_o(\frac{\pi}{2})}{\|e_{ji}\|} & \mathbf{0} & -\frac{b_{jk}^\top \mathcal{R}_o(\frac{\pi}{2})}{\|e_{jk}\|} & \mathbf{0} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}.$$

The right hand side of (8) does not depend on $\sin \alpha_{ijk}$ and is well-defined even if there exist signed angles being 0 or π . It should be noted that the rigidity matrix (8) often has redundant angle information for characterizing infinitesimal rigidity. We provide the following definition.

Definition 5: Given a framework $(\mathcal{G}, p)$, a subset $\bar{\mathcal{T}}_G \subseteq \mathcal{T}_G$ is said to be an *angle index set* (AIS), and the submatrix of (8) with rows corresponding to the derivatives of the signed angles indexed by $\bar{\mathcal{T}}_G$ w.r.t. p , denoted by $R_S^{\bar{\mathcal{T}}_G}(p) \in \mathbb{R}^{|\bar{\mathcal{T}}_G| \times 2n}$, is said to be the corresponding signed angle rigidity matrix.

In bearing rigidity theory, a framework is infinitesimally rigid if and only if the rank of the related rigidity matrix is $2n - 3$. One may question if there is a similar conclusion in signed angle rigidity theory. In [12], the authors proposed the signed angle rigidity theory for angularities and showed that an angularity is infinitesimally signed angle rigid if and only if $\text{rank}(R_S^{\bar{\mathcal{T}}_G}(p)) = 2n - 4$, where an angularity involves a pre-designed AIS $\bar{\mathcal{T}}_G$ specifying constrained signed angles with the assumption that $\sin \alpha_{ijk} \neq 0$ for $(i, j, k) \in \bar{\mathcal{T}}_G$. In

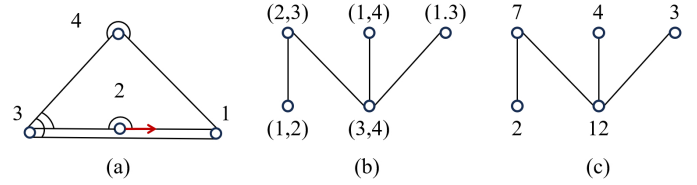


Fig. 2. (a) A non-infinitesimally signed angle rigid framework $(\mathcal{G}, p)$ and an angle connected GAIS $\bar{\mathcal{T}}_G = \{(1, 2, 3), (1, 3, 4), (2, 3, 4), (1, 4, 3)\}$, where the red arrows stand for non-trivial infinitesimal signed angle motions. (b) A graph with the vertices indexing edges in $\mathcal{G}$ and the edges indicating the signed angles w.r.t. $\bar{\mathcal{T}}_G$. (c) The angle index graph of $\mathcal{G}$ and $\bar{\mathcal{T}}_G$, where vertex 12 satisfies $12 = a_{34} = \mathcal{H}(3, 4, 4) = (\min(\{3, 4\}) - 1) \times 4 + \max(\{3, 4\})$.

order to build connections with bearing rigidity theory in a unified architecture, we utilize all the angles in the framework to study the signed angle rigidity property without imposing any assumptions on these angle constraints. That is, given a framework $(\mathcal{G}, p)$, it becomes certain whether there exists an AIS validating (infinitesimal, global) signed angle rigidity.

To better discuss the existence of such a subset, we will propose several definitions in the following subsection.

D. Angle Index Set and Angle Index Graph

Given a framework $(\mathcal{G}, p)$, we present the following definitions regarding the AIS.

Definition 6: For a given framework $(\mathcal{G}, p)$, an AIS $\bar{\mathcal{T}}_G \subseteq \mathcal{T}_G$ is said to be a *global angle index set* (GAIS) if for any $q \in \mathbb{R}^{2n}$ such that $\alpha_{ijk}(q) = \alpha_{ijk}(p)$, $(i, j, k) \in \bar{\mathcal{T}}_G$, it always holds that $S_G(q) = S_G(p)$.

Definition 6 captures the property of an AIS that whether the angles specified by the AIS are sufficient to determine the rest. To provide a geometric intuition, we present an example in Fig. 2 where $\bar{\mathcal{T}}_G$ is a GAIS and $\mathcal{T}_G \setminus \bar{\mathcal{T}}_G = \{(3, 1, 4), (2, 1, 4), (2, 1, 3), (1, 3, 2)\}$. It can be observed that angles α_{134} and α_{143} can determine α_{314} since vertices 1, 4, and 3 form a triangle. Because of $\alpha_{123} = \pi$, vertex 2 can only move between vertices 1 and 3. Hence, $\alpha_{132} = \alpha_{213} = 0$ and $\alpha_{214} = \alpha_{314}$. As a result, the angles indexed by $\mathcal{T}_G \setminus \bar{\mathcal{T}}_G$ can be specified by those indexed by $\bar{\mathcal{T}}_G$.

Definition 7: For a given framework $(\mathcal{G}, p)$, an AIS $\bar{\mathcal{T}}_G \subseteq \mathcal{T}_G$ is said to be a *regular angle index set* (RAIS) if $\text{rank}(R_S^{\bar{\mathcal{T}}_G}(p)) = \text{rank}(R_S^{\mathcal{T}_G}(p))$.

Definition 8: Given a RAIS $\bar{\mathcal{T}}_G$, if there does not exist another RAIS with fewer elements than $\bar{\mathcal{T}}_G$, then it is said to be a *minimal* RAIS. The same goes for the GAIS.

Since an angular constraint is related to two edges, it is necessary to evaluate an AIS from the perspective of the edge set. We use the Hash function $\mathcal{H}(i, j, |\mathcal{V}|) = (\min(\{i, j\}) - 1) \times |\mathcal{V}| + \max(\{i, j\})$ to encode the edge (i, j) of a given graph $\mathcal{G}$ sequentially, and propose the following definitions

Definition 9 (Angle Index Graph): Given a graph $\mathcal{G}$ and an AIS $\bar{\mathcal{T}}_G \subseteq \mathcal{T}_G$, the undirected *angle index graph* is denoted by $\mathcal{G}_A(\bar{\mathcal{T}}_G) \triangleq (\mathcal{V}_A(\mathcal{G}), \mathcal{E}_A(\bar{\mathcal{T}}_G))$, where the vertex set is $\mathcal{V}_A(\mathcal{G}) \triangleq \{a_{ij} = \mathcal{H}(i, j, |\mathcal{V}|) : (i, j) \in \mathcal{E}\}$, $a_{ij} \in \mathcal{V}_A(\mathcal{G})$ is equivalent

to $a_{ji} \in \mathcal{V}_A(\mathcal{G})$, and the edge set is $\mathcal{E}_A(\bar{\mathcal{T}}_{\mathcal{G}}) \triangleq \{(a_{ji}, a_{jk}) : a_{ji}, a_{jk} \in \mathcal{V}_A(\mathcal{G}), (i, j, k) \in \bar{\mathcal{T}}_{\mathcal{G}}\}$.

If $\mathcal{G}$ and $\bar{\mathcal{T}}_{\mathcal{G}}$ are clear from the context, we simply denote $\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{G}}) = (\mathcal{V}_A(\mathcal{G}), \mathcal{E}_A(\bar{\mathcal{T}}_{\mathcal{G}}))$ as $\mathcal{G}_A = (\mathcal{V}_A, \mathcal{E}_A)$.

Definition 10: Given a graph $\mathcal{G}$ and an AIS $\bar{\mathcal{T}}_{\mathcal{G}} \subseteq \mathcal{T}_{\mathcal{G}}$, $\bar{\mathcal{T}}_{\mathcal{G}}$ is said to be *angle connected* if the angle index graph $(\mathcal{V}_A(\mathcal{G}), \mathcal{E}_A(\bar{\mathcal{T}}_{\mathcal{G}}))$ is connected.

Note that $\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{G}}) = (\mathcal{V}_A(\mathcal{G}), \mathcal{E}_A(\bar{\mathcal{T}}_{\mathcal{G}}))$ is formed by mapping the edge $(i, j) \in \mathcal{E}$ to the vertex in $\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{G}})$ and the index $(i, j, k) \in \bar{\mathcal{T}}_{\mathcal{G}}$ to the undirected edge in $\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{G}})$, respectively. For a given graph $\mathcal{G}$, the graph $\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{G}})$ can uniquely deduce an AIS. Fig. 2 illustrates an example of the angle connected set $\bar{\mathcal{T}}_{\mathcal{G}} \subseteq \mathcal{T}_{\mathcal{G}}$, where the angular constraints between edges and their interrelationships are visually presented.

Throughout this paper, we use shorthands SAR, GSAR, and ISAR for signed angle rigid (signed angle rigidity), globally signed angle rigid (global signed angle rigidity), and infinitesimally signed angle rigid (infinitesimal signed angle rigidity), respectively. Similarly, BR, GBR, and IBR are shorthands for corresponding concepts in bearing rigidity theory.

III. GRAPHS FOR SIGNED ANGLE RIGIDITY

In this section, we establish the equivalence between signed angle constraints and bearing constraints in the sense that a framework $(\mathcal{G}, p)$ is IBR, GBR, and BR if and only if it is ISAR, GSAR, and SAR, respectively. Based on these relationships, we derive necessary and sufficient graphical condition for signed angle rigidity.

A. Equivalence on IBR and ISAR

In [12], the authors showed that if $\sin \alpha_{ijk} \neq 0, (i, j, k) \in \mathcal{T}_{\mathcal{G}}$, the null space of the signed angle rigidity matrix $R_S^{\mathcal{T}_{\mathcal{G}}}(p)$ contains the following three linear spaces:

$$\mathcal{M}_s \triangleq \text{span}\{p\}, \quad (9)$$

$$\mathcal{M}_r \triangleq \text{span}\left\{[I_n \otimes \mathcal{R}_o(\frac{\pi}{2})]p\right\}, \quad (10)$$

$$\mathcal{M}_t \triangleq \text{span}\{[I_n \otimes [1, 0]^T, \mathbf{1}_n \otimes [0, 1]^T]\}. \quad (11)$$

Note that any uniform rotation of p can be uniquely expressed as the linear combination of p and $[I_n \otimes \mathcal{R}_o(\frac{\pi}{2})]p$. Moreover, according to Lemma 1, it can be verified that the result remains valid even when there exist triplets $(i, j, k) \in \mathcal{T}_{\mathcal{G}}$ satisfying $\sin \alpha_{ijk} = 0$. Therefore, the following lemma can be derived from Definition 4 naturally, which implies that the trivial infinitesimal signed angle motions of a framework in $\mathbb{R}^2$ include 2-dimensional uniform translations, 1-dimensional uniform scalings, and 1-dimensional uniform rotations.

Lemma 2: In $\mathbb{R}^2$, a framework $(\mathcal{G}, p)$ is ISAR if and only if $\text{Null}(R_S^{\mathcal{T}_{\mathcal{G}}}(p)) = \mathcal{M}_s \cup \mathcal{M}_r \cup \mathcal{M}_t$.

As shown in Fig. 2(a), the framework is not ISAR since there exists a non-trivial infinitesimal signed angle motion. Lemma 2 also indicates that a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$ is ISAR if and only if $\text{rank}(R_S^{\mathcal{T}_{\mathcal{G}}}(p)) = 2n - 4$.

Theorem 1: In $\mathbb{R}^2$, a framework $(\mathcal{G}, p)$ is ISAR if and only if it is IBR.

Proof: See Appendix VIII-A. ■

Theorem 1 implies that the conditions on the framework for bearing constraints and signed angle constraints to exclude non-trivial infinitesimal motions are the same. In [10], it has been shown that the shape of an IBR framework can always be uniquely fixed by all bearings among neighboring nodes. However, Theorem 1 does not imply an ISAR framework can fix its shape by using all angles. To seek the conditions on frameworks for signed angles to constrain a shape uniquely, we next study the relationship between GSAR and GBR.

B. Equivalence on GSAR and GBR

In this subsection, we establish the equivalence among BR, SAR, GBR, and GSAR. Before that, we propose a graphical condition, under which signed angle constraints are equivalent to bearing constraints regardless of uniform rotations.

Lemma 3: Given a connected graph $\mathcal{G}$ and two configurations $p, q \in \mathbb{R}^{2n}$, $S_{\mathcal{G}}(q) = S_{\mathcal{G}}(p)$ if and only if $B_{\mathcal{G}}(q) = [I_{|\mathcal{E}|} \otimes \mathcal{R}_o(\theta)] B_{\mathcal{G}}(p)$ for some $\theta \in [0, 2\pi)$.

Proof: See Appendix VIII-B. ■

Although Lemma 3 is not difficult to achieve, it reveals a non-trivial fact: despite being measured in different local coordinate frames, signed angle constraints are able to function as global bearing constraints in a certain coordinate frame as long as the graph is connected. This interesting property implies the great potential of signed angles in coordinate-free network localization and formation control, but surprisingly, is not found in the literature. Now we are ready to show the equivalence between GBR and GSAR.

Theorem 2: A framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$ is BR (GBR) if and only if it is SAR (GSAR).

Proof: See Appendix VIII-C. ■

Combining Theorems 1 and 2, together with [10, Theorem 3] and [10, Theorem 5], the following results hold.

Corollary 1: If a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$ is ISAR, then it is GSAR.

Corollary 2: A framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$ is GSAR if and only if it is SAR.

To clearly illustrate the results stated above, we summarize the relationships between different concepts of bearing rigidity and signed angle rigidity in Fig. 3.

Remark 1: By pre-specifying the set of angle constraints, the authors in [33] provided some necessary conditions for rigidity. In [12, 34], the authors further systematically investigated the rigidity theory under pre-designed angle constraints and showed that rigidity is not equivalent to global rigidity, and infinitesimal rigidity does not imply global rigidity. In this paper, we derive the necessary and sufficient conditions for signed angle rigidity and arrive at different conclusions. This is because we utilize all the subtended angles in frameworks to characterize the rigidity, instead of partial angles. Therefore,

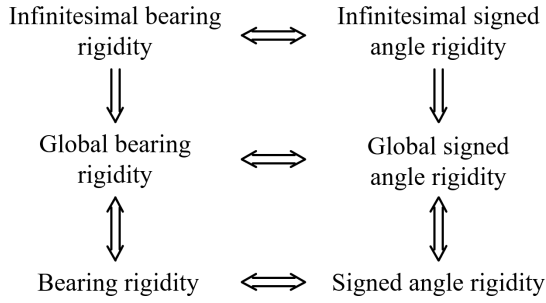


Fig. 3. A relation conclusion for different concepts in bearing rigidity theory and signed angle rigidity theory in 2-D

the rigidity property is fully dependent on the framework rather than the angle constraint set, this benefits for connecting the rigidity property with graph structures.

Remark 2: We have established complete equivalence on different rigidity properties based on signed angle constraints and bearing constraints. One may question if such equivalence holds for unsigned angles in [11]. Similarly to the conclusion of shape fixability in bearing rigidity theory [10], it has been proven in [17] that a framework in $\mathbb{R}^2$ can be uniquely determined by unsigned angles if and only if it is infinitesimally angle rigid and globally angle rigid. Although several illustrations induce the conjecture that infinitesimal angle rigidity may imply global angle rigidity (like ISAR imply GSAR), this is yet to be proven.

C. The Uniqueness of Shape

Based on the above results, we next answer an important question: under what conditions can the shape of a framework be uniquely determined by signed angle constraints?

Given a configuration $p \in \mathbb{R}^{2n}$, the set of configurations that form the same shape as p can be described as:

$$\mathcal{E}(p) = \{q \in \mathbb{R}^{2n} : q = c[I_n \otimes \mathcal{R}_o(\theta)]p + \mathbf{1}_n \otimes \xi, \theta \in [0, 2\pi), c \in \mathbb{R}^+, \xi \in \mathbb{R}^2\}, \quad (12)$$

where ξ , θ , and c are the translational, rotational, and scaling factor, respectively. If $p \in \mathbb{R}^{2n}$ is non-degenerate, then $\mathcal{E}(p)$ is a 4-dimensional smooth manifold. Given a graph $\mathcal{G}$ with n vertices, the configuration set of frameworks with the same signed angles as $(\mathcal{G}, p)$ can be depicted as:

$$\mathcal{E}_{\mathcal{G}}(p) = \{q \in \mathbb{R}^{2n} : \alpha_{ijk}(q) = \alpha_{ijk}(p), (i, j, k) \in \mathcal{T}_{\mathcal{G}}\}. \quad (13)$$

Lemma 4: The shape of a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$ can be uniquely determined by signed angles up to uniform rotations, translations, and scalings if and only if $\mathcal{E}(p) = \mathcal{E}_{\mathcal{G}}(p)$.

The following theorem presents the necessary and sufficient conditions for the shape fixability of frameworks under signed angle constraints, from the perspectives of ISAR and GSAR, respectively.

Theorem 3: Given a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, the following statements are equivalent:

- (i) $(\mathcal{G}, p)$ is ISAR;
- (ii) $(\mathcal{G}, p)$ is non-degenerate and GSAR;
- (iii) the shape of $(\mathcal{G}, p)$ can be uniquely determined by signed angles up to the translational, rotational, and scaling factor.

Proof: See Appendix VIII-D. ■

Given a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, Theorem 3 suggests that: (i) ISAR, which can be examined by computing the rank of the signed angle rigidity matrix, is able to determine whether the shape of a framework can be specified by signed angles uniquely; (ii) ISAR (IBR) and GSAR (GBR) imply each other when p is non-degenerate.

However, given the above results, computing the rank of the rigidity matrix for large-scale networks may be computationally prohibitive. In the next subsection, we study how to efficiently construct graphs that are ISAR (GSAR, SAR) for almost all configurations.

D. Signed Angle Rigid Graphs

To begin with, we formally introduce the definition of signed angle rigidity for graphs.

Definition 11: A graph is said to be SAR (ISAR, GSAR) if there exists at least one configuration $p \in \mathbb{R}^{2n}$ such that $(\mathcal{G}, p)$ is SAR (ISAR, GSAR).

In [27, Lemma 2], IBR has been shown as a generic property of graphs. More specifically, if $(\mathcal{G}, p)$ is IBR for a configuration $p \in \mathbb{R}^{2n}$, then $(\mathcal{G}, q)$ is IBR for almost all $q \in \mathbb{R}^{2n}$. According to [10], such a graphic property holds for GBR and BR as well. From the relationship between bearing rigidity and signed angle rigidity shown in Fig. 3, it can be observed that ISAR, GSAR, and SAR are essentially determined by the graph rather than the configuration, thus are generic properties of graphs. We summarize this result in the following theorem.

Theorem 4: If a graph $\mathcal{G}$ is ISAR (GSAR, SAR), then the framework $(\mathcal{G}, p)$ is ISAR (GSAR, SAR) for almost all configurations $p \in \mathbb{R}^{2n}$ in the sense that the set of $q \in \mathbb{R}^{2n}$ such that $(\mathcal{G}, q)$ is not ISAR (GSAR, SAR) is of measure zero.

The equivalent graphical description for bearing rigidity has been well understood in [27, 36]. Inspired by this, we derive the equivalent graphical condition for signed angle rigidity. Before that, we introduce a class of well-known graphs.

Definition 12 (Laman Graph [27, 28]): A n -point ($n \geq 3$) graph $\mathcal{L}_n = (\mathcal{V}_n, \mathcal{E}_n)$ satisfying $|\mathcal{E}_n| = 2n - 3$ is a Laman graph if and only if $e \leq 2v - 3$ for every subgraph of $\mathcal{L}_n$ having v vertices and e edges.

Combining Theorem 1 and [27, Theorem 2], we present the following result:

Theorem 5: A graph $\mathcal{G}$ is ISAR if and only if it contains a Laman spanning subgraph.

In [36], the authors have shown that minimally IBR graphs are Laman graphs. However, they did not obtain a similar conclusion for minimally BR graphs. According to Theorem

3 and Theorem 4, ISAR and GSAR imply each other. As a result, a graph $\mathcal{G}$ is GSAR if and only if it contains a Laman spanning subgraph. Based on the equivalence between bearing rigidity and signed angle rigidity, together with [36, Theorem 4.1], we have the following theorem.

Theorem 6: A graph $\mathcal{G}$ is minimally SAR (ISAR, GSAR, IBR, GBR, BR) if and only if it is a Laman graph.

Now, it is clear which type of frameworks can be uniquely determined by signed angles. In order to exclude redundant angle constraints, the problem of finding a minimal set of signed angle constraints for shape determination will be investigated in the next section.

Remark 3: Combining Theorem 6 and [10, Theorem 8], we can obtain that ISAR is equivalent to infinitesimal distance rigidity (IDR). Note that given an IDR framework in $\mathbb{R}^2$, the $2n-3$ linearly independent rows of the distance rigidity matrix always correspond to the edges of an IDR Laman spanning subframework [28, Lemma 2.3]. Therefore, for any ISAR framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, we can find an ISAR Laman spanning subframework in $(\mathcal{G}, p)$ by finding linearly independent rows of the distance rigidity matrix.

IV. ANGLE INDEX SETS FOR SIGNED ANGLE RIGIDITY

Finding a minimal set of angle constraints is an important problem, and has been investigated in [11, 12, 34]. However, these results are limited to conservative graph structures. In this section, taking the advantage of the general graphical condition for signed angle rigidity established in Section III, we will propose an approach to construction of a minimal GAIS/RAIS for any ISAR frameworks.

A. Minimal GAISs for Signed Angle Rigidity

Given a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$ and an AIS $\bar{\mathcal{T}}_{\mathcal{G}} \subseteq \mathcal{T}_{\mathcal{G}}$, the configuration set of frameworks with the same signed angles indexed by $\bar{\mathcal{T}}_{\mathcal{G}}$ as $(\mathcal{G}, p)$ can be depicted as:

$$\mathcal{E}_{\bar{\mathcal{T}}_{\mathcal{G}}}(p) = \{q \in \mathbb{R}^{2n} : \alpha_{ijk}(q) = \alpha_{ijk}(p), (i, j, k) \in \bar{\mathcal{T}}_{\mathcal{G}}\}, \quad (14)$$

From Definition 6, we have the following lemma naturally.

Lemma 5: For a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, $\bar{\mathcal{T}}_{\mathcal{G}} \subseteq \mathcal{T}_{\mathcal{G}}$ is a GAIS if and only if it satisfies $\mathcal{E}_{\bar{\mathcal{T}}_{\mathcal{G}}}(p) = \mathcal{E}_{\mathcal{G}}(p)$.

The following result provides an intuitive condition to determine GAISs. Since it is similar to that of Lemma 3, we omit its proof here.

Lemma 6: For a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, an AIS $\bar{\mathcal{T}}_{\mathcal{G}}$ is a GAIS if it is angle connected.

It is worth noting that the angle connectivity of GAISs is not necessary, see counterexamples in Fig. 4. Interestingly, when the framework is embedded by a Laman graph (termed as a *Laman framework*), one has a different conclusion.

Theorem 7: For an ISAR Laman framework $(\mathcal{L}_n, p)$ in $\mathbb{R}^2$, $\bar{\mathcal{T}}_{\mathcal{L}_n} \subseteq \mathcal{T}_{\mathcal{L}_n}$ is a minimal GAIS if and only if $|\bar{\mathcal{T}}_{\mathcal{L}_n}| = 2n - 4$ and $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is angle connected.

Proof: See Appendix VIII-E. ■

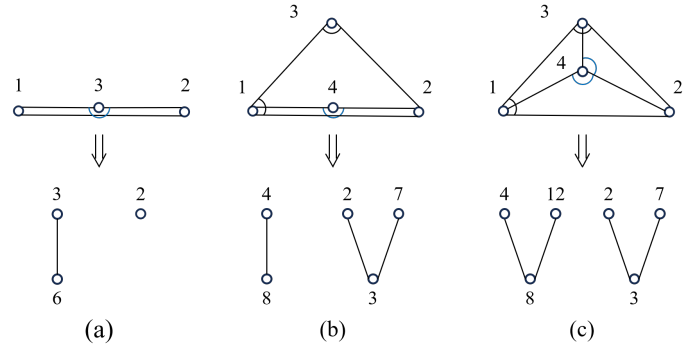


Fig. 4. Examples of GAISs whose corresponding angle index graphs are not angle connected. Black and blue are used to distinguish different components of $\bar{\mathcal{T}}_{\mathcal{G}}$. (a) A GSAR framework with a GAIS $\bar{\mathcal{T}}_{\mathcal{G}} = \{(1, 3, 2)\}$. (b) A non-GSAR framework with a GAIS $\bar{\mathcal{T}}_{\mathcal{G}} = \{(1, 4, 2), (1, 3, 2), (2, 1, 3)\}$. (c) A GSAR framework with a GAIS $\bar{\mathcal{T}}_{\mathcal{G}} = \{(1, 4, 2), (2, 4, 3), (1, 3, 2), (2, 1, 3)\}$.

On the one hand, according to the proof of Theorem 7, we observe that for a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, the breadth first search method (BFS) [37, Section 20.2] can be applied to extract a spanning tree from the angle index graph $\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{G}}) = (\mathcal{V}_A(\mathcal{G}), \mathcal{E}_A(\bar{\mathcal{T}}_{\mathcal{G}}))$. The resulting AIS $\bar{\mathcal{T}}_{\mathcal{G}}$ associated with this spanning tree is angle connected and satisfies $|\bar{\mathcal{T}}_{\mathcal{G}}| = |\mathcal{E}| - 1$. On the other hand, according to Theorems 3-5, a framework can be uniquely determined by signed angles if and only if it contains an ISAR Laman subframework. Based on these observations, we propose Algorithm 1 to construct a minimal GAIS $\bar{\mathcal{T}}_{\mathcal{G}}$ for arbitrary ISAR frameworks, which enables unique shape while satisfying $|\bar{\mathcal{T}}_{\mathcal{G}}| = 2n - 4$.

Algorithm 1 Finding a Minimal GAIS $\bar{\mathcal{T}}_{\mathcal{G}}$ for Any ISAR Framework $(\mathcal{G}, p)$

Input: An ISAR framework $(\mathcal{G}, p)$

Output: $\bar{\mathcal{T}}_{\mathcal{G}}$

- 1: Find an ISAR Laman spanning subframework $(\mathcal{L}_n, p)$ from $(\mathcal{G}, p)$ according to Remark 3. Let $\bar{\mathcal{T}}_{\mathcal{G}} \leftarrow \emptyset$;
- 2: Find a spanning tree $(\mathcal{V}_A^{tree}, \mathcal{E}_A^{tree})$ within the angle index graph $(\mathcal{V}_A(\mathcal{L}_n), \mathcal{E}_A(\mathcal{T}_{\mathcal{L}_n}))$ by BFS [37, Section 20.2].
- 3: **for all** $(a_{ji}, a_{jk}) \in \mathcal{E}_A^{tree}$ **do**
- 4: Let $\bar{\mathcal{T}}_{\mathcal{G}} \leftarrow \{(i, j, k)\} \cup \bar{\mathcal{T}}_{\mathcal{G}}$ if $i < k$, $\bar{\mathcal{T}}_{\mathcal{G}} \leftarrow \{(k, j, i)\} \cup \bar{\mathcal{T}}_{\mathcal{G}}$ otherwise;
- 5: **end for**
- 6: **return** $\bar{\mathcal{T}}_{\mathcal{G}}$

Remark 4: Algorithm 1 is a polynomial-time algorithm. As noted in Remark 3, Line 1 can be achieved via Gaussian elimination [38, Section 3.2], with a time complexity of $\mathcal{O}(|\mathcal{E}|n^2)$ when $|\mathcal{E}| \geq 2n$, and $\mathcal{O}(|\mathcal{E}|^2n)$ otherwise. Since the worst-case time complexity of Lines 2–5 is $\mathcal{O}(n^2)$ (when $(\mathcal{V}_A(\mathcal{L}_n), \mathcal{E}_A(\mathcal{T}_{\mathcal{L}_n}))$ is a complete graph) and $|\mathcal{E}| \leq n(n-1)/2$, the worst-case time complexity of Algorithm 1 is $\mathcal{O}(n^5)$.

B. Minimal RAISs for Signed Angle Rigidity

The following result provides an intuitive condition to determine RAISs.

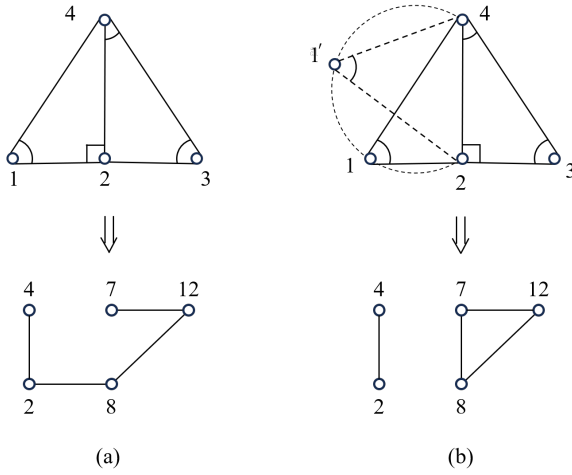


Fig. 5. Examples of angle connected (non-angle-connected) AISs and their angle index graphs, where the framework is embedded by a Laman graph $\mathcal{L}_4$ and satisfies ISAR under the set $\mathcal{T}_{\mathcal{L}_n}$. The shape of (a) can be determined by angle constraints due to the angle connectivity of the related set $\bar{\mathcal{T}}_{\mathcal{L}_4}$. (b) is flexible since the angle constraints α_{324} , α_{243} , α_{234} are linearly dependent.

Lemma 7: For a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, an AIS $\bar{\mathcal{T}}_{\mathcal{G}}$ is a RAIS if it is angle connected.

Proof: See Appendix VIII-F. ■

Similar to GAISs, the angle connectivity of RAISs is not necessary, see an example in Fig. 4(a), where $\bar{\mathcal{T}}_{\mathcal{G}}$ is a RAIS since $\text{rank}(R_S^{\bar{\mathcal{T}}_{\mathcal{G}}}(p)) = \text{rank}(R_S^{\mathcal{T}_{\mathcal{G}}}(p)) = 1 < 2$ but not angle connected. Nonetheless, we can have the following conclusion.

Theorem 8: For an ISAR Laman framework $(\mathcal{L}_n, p)$ in $\mathbb{R}^2$, the following statements are equivalent:

- (i) $\bar{\mathcal{T}}_{\mathcal{L}_n} \subseteq \mathcal{T}_{\mathcal{L}_n}$ is a minimal GAIS;
- (ii) $\bar{\mathcal{T}}_{\mathcal{L}_n} \subseteq \mathcal{T}_{\mathcal{L}_n}$ is a minimal RAIS;
- (iii) $|\bar{\mathcal{T}}_{\mathcal{L}_n}| = 2n - 4$ and $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is angle connected.

Proof: See Appendix VIII-G. ■

Theorem 8 indicates that the AIS generated by Algorithm 1 is a minimal RAIS (GAIS) and the angle connectivity of this set guarantees the independence of constraints. Since $|\mathcal{E}_n| = 2n - 3$, there always exists an angle connected AIS $\bar{\mathcal{T}}_{\mathcal{L}_n}$ with $|\bar{\mathcal{T}}_{\mathcal{L}_n}| = |\mathcal{E}_n| - 1$ whose angle index graph related to $\mathcal{L}_n$ is a spanning tree of $\mathcal{G}_A(\mathcal{T}_{\mathcal{L}_n})$, and the constraints indexed by the set $\bar{\mathcal{T}}_{\mathcal{L}_n}$ can specify the framework and support $\text{rank}(R_S^{\bar{\mathcal{T}}_{\mathcal{L}_n}}(p)) = 2n - 4$. If the AIS with $|\bar{\mathcal{T}}_{\mathcal{L}_n}| = 2n - 4$ is not angle connected, then there must exist a cycle $\mathcal{C} = (\mathcal{V}_C, \mathcal{E}_C)$ in the angle index graph $\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{L}_n})$ and the signed angle constraints corresponding to the edges of $\mathcal{C}$ are linearly dependent. As a result, their time derivatives are also linearly dependent, which indicates $\text{rank}(R_S^{\bar{\mathcal{T}}_{\mathcal{L}_n}}(p)) < 2n - 4$, see Figure 5 for an illustration.

To demonstrate the results stated above, we summarize the relationship between AISs and signed angle rigidity in Fig. 6.

V. SIGNED ANGLE-BASED SENSOR NETWORK LOCALIZATION

In this section, we study the problem of signed angle-based sensor network localization. Based on the established signed

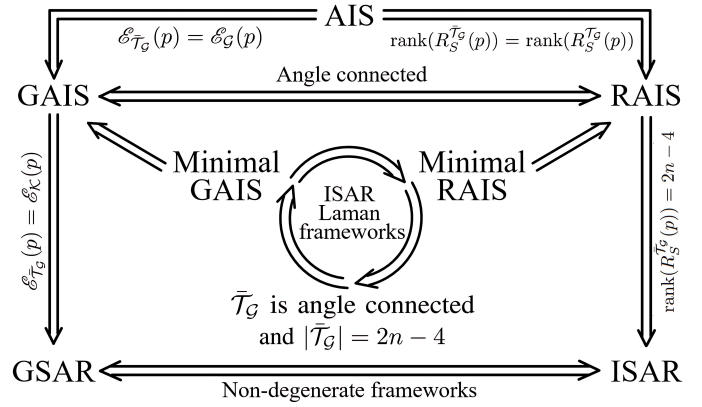


Fig. 6. A graph illustrating the relationship between different concepts related to AISs and signed angle rigidity properties. An arrow means ‘implication’. Specifically, ‘ $A \Rightarrow B$ with a condition noted on the arrow’ means that A implies B when the condition holds.

angle rigidity theory, we will propose a distributed localization algorithm with global convergence guarantee.

A. Problem Formulation and Network Localizability

Given a sensor network, the sensors can be represented as $\mathcal{V} = \mathcal{A} \cup \mathcal{F}$ where the sensors in $\mathcal{A} \subseteq \mathcal{V}$ are called *anchors* whose locations are known, and the sensors in $\mathcal{F} = \mathcal{V} \setminus \mathcal{A}$ are called *followers* whose locations are to be determined. Furthermore, the real location of the network is denoted by $p = [p_1, \dots, p_n]^\top$, where $p_i \neq p_j$ for any pair of sensors $i, j \in \mathcal{V}$. The interactions between the sensors are indicated by the fixed and undirected sensing graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

We use $(\mathcal{G}, p, \mathcal{A})$ to describe a sensor network and consider the localization problem as follows.

Problem 1: Consider a sensor network $(\mathcal{G}, p, \mathcal{A})$ in $\mathbb{R}^2$, design a distributed algorithm for each follower $i \in \mathcal{F}$ to estimate its location p_i merely using angle measurements $\{\alpha_{ijk}\}_{(i,j,k) \in \mathcal{T}_{\mathcal{G}}}$ and communicated information from neighbors, i.e., seek a solution to the following nonlinear equations,

$$\begin{cases} \frac{\hat{p}_j - \hat{p}_k}{\|\hat{p}_j - \hat{p}_k\|} = \mathcal{R}_o(\alpha_{ijk}) \frac{\hat{p}_j - \hat{p}_i}{\|\hat{p}_j - \hat{p}_i\|} & \forall (i, j, k) \in \mathcal{T}_{\mathcal{G}}, \\ \hat{p}_i = p_i & i \in \mathcal{A}, \end{cases} \quad (15)$$

where $\hat{p}_i$ is the estimated location of follower i .

The presence of anchors with known positions has also been considered in the literature, e.g., [18, 39–42]. Compared with them, the required topology for effectiveness of the algorithm will be relaxed in this paper. Noted that (15) may have multiple solutions, not all of which correspond to the real location of the network. Therefore, it is necessary to investigate the condition for the existence of a unique solution to (15), which motivates the introduction of the “localizability” notion.

Definition 13: A sensor network $(\mathcal{G}, p, \mathcal{A})$ in $\mathbb{R}^2$ is said to be *signed angle localizable* if the real location p is the only feasible solution to (15).

The following theorem proposes a necessary and sufficient condition for signed angle localizability.

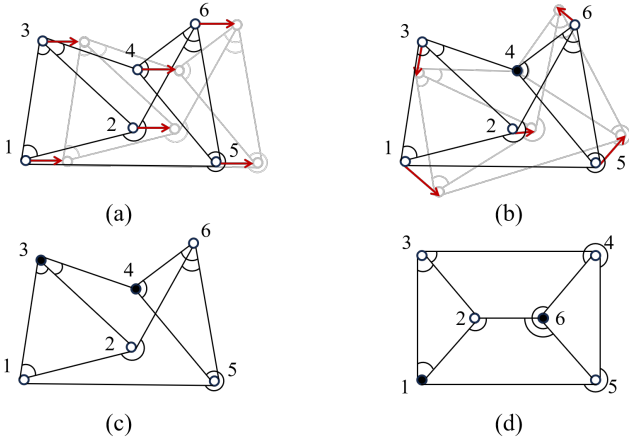


Fig. 7. Examples of sensor networks. Anchors and followers are denoted by solid and hollow dots, respectively. The red arrows represent the trivial infinitesimal signed angle motions related to the followers. (a) and (b) are not localizable due to the existence of uniform translation and rotation motions. (c) is signed angle localizable since it is characterized by an ISAR Laman framework and the condition $|\mathcal{A}| \geq 2$ specifies the rotational, translational, and scaling factor. Although the framework w.r.t. (d) is not ISAR since vertices 4, 5, 6 can simultaneously translate along the bearing b_{34} while preserving all signed angle constraints, fixing the locations of vertices 1 and 6 ensures invariance of the distance between them. Hence, nontrivial infinitesimal motions are eliminated and the network in Figure 7(d) is signed angle localizable.

Theorem 9: For an ISAR framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, the sensor network $(\mathcal{G}, p, \mathcal{A})$ is signed angle localizable if and only if $|\mathcal{A}| \geq 2$.

Proof: See Appendix VIII-H. ■

Examples in Figs. 7(a)-(b) show non-localizable networks that violate the condition of Theorem 9. Fig. 7(c) shows a signed angle localizable network, which can be verified by Theorem 9. In contrast, the localizability conditions in [18] only apply to trigraphs and thus is inapplicable in this case. Additionally, it can be observed from Fig. 7(d) that the ISAR of the framework is not necessary to guarantee signed angle localizability since additional position constraints of anchors may eliminate the existence of undesired solutions for (15).

Remark 5: Problem 1 involves all signed angles in the sensor network. However, when $(\mathcal{G}, p)$ is ISAR, only partial signed angles are required. That is, $\bar{\mathcal{T}}_{\mathcal{G}}$ can be replaced by the minimal GAIS $\bar{\mathcal{T}}_{\mathcal{G}}$ induced by Algorithm 1. When the scenario does not support the centralized implementation of Algorithm 1, each sensor $i \in \mathcal{V}$ can generate a local AIS ${}^i\bar{\mathcal{T}}_{\mathcal{G}}$ such that the angle index graph $(\mathcal{V}_A(\bar{\mathcal{G}}_i), \mathcal{E}_A({}^i\bar{\mathcal{T}}_{\mathcal{G}}))$ is connected and $|{}^i\bar{\mathcal{T}}_{\mathcal{G}}| = |\mathcal{N}_{\mathcal{G}}^i| - 1$, where $\bar{\mathcal{G}}_i = (\bar{\mathcal{V}}_i, \bar{\mathcal{E}}_i)$, $\bar{\mathcal{V}}_i = \mathcal{N}_{\mathcal{G}}^i \cup \{i\}$, and $\bar{\mathcal{E}}_i = \{(i, j) \in \mathcal{E} : j \in \mathcal{N}_{\mathcal{G}}^i\}$. Then, the angle index set of the network is $\bar{\mathcal{T}}_{\mathcal{G}} = \bigcup_{i \in \mathcal{V}} {}^i\bar{\mathcal{T}}_{\mathcal{G}}$ with $|\bar{\mathcal{T}}_{\mathcal{G}}| = \sum_{i \in \mathcal{V}} (|\mathcal{N}_{\mathcal{G}}^i| - 1) = 2|\mathcal{E}| - n$. It can be verified that $\bar{\mathcal{T}}_{\mathcal{G}}$ is angle connected and sufficient for localization according to Lemma 6. However, $\bar{\mathcal{T}}_{\mathcal{G}}$ is not minimal even if $(\mathcal{G}, p)$ is minimally ISAR (i.e., $\mathcal{G}$ is a Laman graph) since $|\bar{\mathcal{T}}_{\mathcal{G}}| \geq 2(2n - 3) - n = 3n - 6 > 2n - 4$ when $n > 2$.

B. A Distributed Localization Algorithm

Next, we design a distributed algorithm to achieve the localization objective under the following assumption.

Assumption 1: The sensor network $(\mathcal{G}, p, \mathcal{A})$ considered satisfies: (i) $(\mathcal{G}, p)$ is an ISAR Laman framework; (ii) $|\mathcal{A}| \geq 2$; (iii) there exist two anchors $i, j \in \mathcal{A}$ satisfying $(i, j) \in \mathcal{E}$.

In what follows, we consider that each sensor $i \in \mathcal{V}$ has access to angle measurements associated with the AIS

$${}^i\bar{\mathcal{T}}_{\mathcal{G}} = \{(k, i, j) \in \mathcal{V}^3 : (k, i, j) \in \bar{\mathcal{T}}_{\mathcal{G}}\} \quad (16)$$

where $\bar{\mathcal{T}}_{\mathcal{G}}$ is induced by Algorithm 1 under Assumption 1.

Inspired by [16], each follower $i \in \mathcal{F}$ generates a random initial value $\hat{p}_i(0) \in \mathbb{R}^2$ and updates its estimated location by

$$\dot{\hat{p}}_i(t) = - \sum_{j \in \mathcal{N}_i} P(\hat{b}_{ij}(t))(\hat{p}_i(t) - \hat{p}_j(t)), \quad (17)$$

where $\hat{b}_{ij}(t) \in \mathbb{R}^2$ is the estimate of the global bearing $b_{ij} \triangleq b_{ij}(p)$ and updated through the estimator (18) and independent of the position estimates $\hat{p}_i(t)$ for all $i \in \mathcal{V}$ and $t \geq 0$, $\hat{p}_j(t)$ is obtained via communications with agent $j \in \mathcal{N}_i$. (In the case of anchors, $\hat{p}_i(t) = p_i$ for all time $t \geq 0$.)

To estimate the global bearings required for the updating law (17), each sensor $i \in \mathcal{V}$ generates a random initial estimate $\hat{b}_{ij}(0) \in \mathbb{R}^2$ for each $j \in \mathcal{N}_i$ and updates $\hat{b}_{ij}(t)$ by

$$\begin{aligned} \dot{\hat{b}}_{ij}(t) = & - \left\{ \sum_{(j, i, k_1) \in \bar{\mathcal{T}}_{\mathcal{G}}} \left[\hat{b}_{ij}(t) - \mathcal{R}_o^\top(\alpha_{jik_1}) \hat{b}_{ik_1}(t) \right] \right. \\ & + \sum_{(k_2, i, j) \in \bar{\mathcal{T}}_{\mathcal{G}}} \left[\hat{b}_{ij}(t) - \mathcal{R}_o(\alpha_{k_2ij}) \hat{b}_{ik_2}(t) \right] \\ & + \sum_{(i, j, k_3) \in \bar{\mathcal{T}}_{\mathcal{G}}} \left[\hat{b}_{ij}(t) + \mathcal{R}_o^\top(\alpha_{ijk_3}) \hat{b}_{jk_3}(t) \right] \\ & \left. + \sum_{(k_4, j, i) \in \bar{\mathcal{T}}_{\mathcal{G}}} \left[\hat{b}_{ij}(t) + \mathcal{R}_o(\alpha_{k_4ji}) \hat{b}_{jk_4}(t) \right] \right\}, \quad (18) \end{aligned}$$

where α_{jik_1} and α_{k_2ij} are measured by sensor i , $\hat{b}_{jk_3}(t)$, $\hat{b}_{jk_4}(t)$, α_{ijk_3} , and α_{k_4ji} are obtained via communications with agent $j \in \mathcal{N}_i$. (In the case of $(i, j) \in \mathcal{E}$ and $i, j \in \mathcal{A}$, $\hat{b}_{ij}(t) = b_{ij}$ and $\dot{\hat{b}}_{ij}(t) = \mathbf{0}$ for all $t \geq 0$, where b_{ij} is computed from the locations of anchors i and j .)

Note that under the action of (18), $\hat{b}_{ij}(t)$ may not satisfy $\hat{b}_{ij}(t) = \hat{b}_{ji}(t)$ and $\|\hat{b}_{ij}(t)\| = 1$ for all $t \geq 0$. Hence, $P(\hat{b}_{ij}(t))$ may not be an orthogonal projection for $t \geq 0$. However, these deviations do not affect the convergence of $\hat{p}_i(t)$ to p_i for all $i \in \mathcal{F}$.

Theorem 10 (Global Convergence): Under Assumption 1 and the AIS (16), by implementing the algorithm (17)–(18), $\hat{p}_i(t)$ converges to p_i exponentially under arbitrary initial estimates $\hat{p}_i(0)$ and $\{\hat{b}_{ij}(0)\}_{j \in \mathcal{N}_i}$ for all $i \in \mathcal{V}$.

Proof: See Appendix VIII-I. ■

Although the proposed localization protocol is similar to [43], it does not rely on orientation estimation to get global bearings. Compared with the localization algorithms based

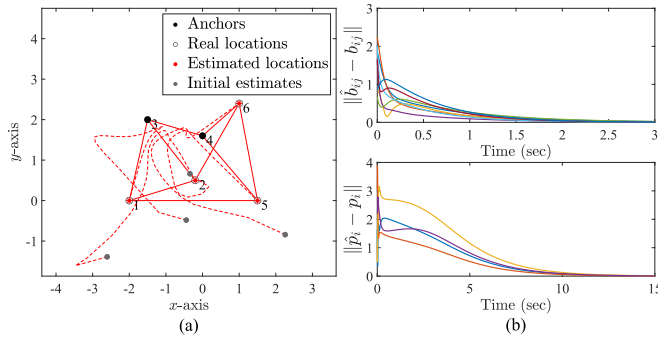


Fig. 8. (a) Trajectories of the estimation of network locations from randomly initial states to real positions. (b) Evolution of bearing and location estimation errors.

on local bearing measurements [18, 39–42], the distributed protocol (17)–(18) requires milder conditions for the network topology. For example, the sensor network in Fig. 7(c) does not satisfy the condition required in [18, 39–42], but is sufficient for our localization algorithm, see Example 1.

Example 1: Consider a six-sensor network $(\mathcal{G}, p, \mathcal{A})$ described in Fig. 7(c), which satisfies Assumption 1. The initial estimates $\hat{p}_i(0)$ and $\{\hat{b}_{ij}(0)\}_{j \in \mathcal{N}_i}$ are randomly generated for each sensor $i \in \mathcal{V}$. By implementing (17)–(18), the convergence trajectories are shown in Fig. 8(a), and the evolutions of bearing and location estimation errors are shown, respectively, in Fig. 8(b). From the simulation results, we observe that the estimated locations converge to real locations asymptotically, which is consistent with Theorem 10.

Remark 6: The first condition in Assumption 1 can be relaxed to ISAR frameworks. When the framework $(\mathcal{G}, p)$ is ISAR but not a Laman framework, we can first implement Algorithm 1 to induce a minimal GAIS $\bar{\mathcal{T}}_{\mathcal{G}}$. If $\bar{\mathcal{T}}_{\mathcal{G}}$ does not contain a triple (i, j, k) (or (k, j, i)) satisfying $(i, j) \in \mathcal{E}$ and $i, j \in \mathcal{A}$, we extend $\bar{\mathcal{T}}_{\mathcal{G}}$ to include such a triple (i, j, k) (or (k, j, i)) for some $k \in \mathcal{N}_j$ and $i, j \in \mathcal{A}$. Then, each sensor $i \in \mathcal{V}$ measures the angles indexed by (16), and communicates with neighbors $j \in \bar{\mathcal{N}}_i$ to update the estimates $\hat{b}_{ij}(t)$ and $\hat{p}_i(t)$, where $\bar{\mathcal{N}}_i = \{j, k \in \mathcal{V} : \exists (i, j, k), (k, i, j), \text{ or } (j, k, i) \in \bar{\mathcal{T}}_{\mathcal{G}}\}$. That is, the subscript “ $j \in \mathcal{N}_i$ ” in (17) is replaced by “ $j \in \bar{\mathcal{N}}_i$ ”. It can be verified that $\hat{p}_i(t)$ globally converges to p_i for all $i \in \mathcal{F}$ under the distributed protocol (17)–(18) when $(\mathcal{G}, p)$ is ISAR but not a Laman framework.

VI. SIGNED ANGLE RIGID FORMATION CONTROL

In this section, we will demonstrate the application of the developed rigidity theory to formation control, where communications between agents are disabled. The target formation is described by signed angle constraints and each agent in the formation only measures signed angles to meet these constraints via a distributed control law.

For the group of n agents moving in the plane, we use $\mathcal{V} = \{1, \dots, n\}$ to label the agents. Let $p_i \triangleq p_i(t) \in \mathbb{R}^2$ be the position of agent $i \in \mathcal{V}$, $\beta_i \triangleq \beta_i(t) \in [0, 2\pi)$ be the signed

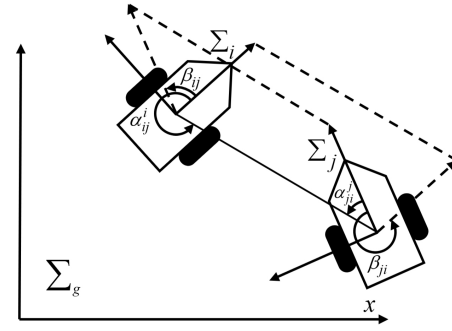


Fig. 9. The illustration of the measured values required for the controllers.

angle from the body frame Σ_i to the coordinate frame Σ_g , $p \triangleq [p_1^\top, \dots, p_n^\top]^\top$, and $\beta \triangleq [\beta_1, \dots, \beta_n]^\top$. The position and attitude dynamics of agent i is

$$\dot{p}_i = \mathcal{R}_o(\beta_i) u_i^p(t), \quad (19)$$

$$\dot{\beta}_i = u_i^a(t), \quad (20)$$

where $u_i^p(t)$ and $u_i^a(t)$ are the control input to be designed.

In this section, if $j \in \mathcal{N}_i$, agent i can measure the signed angle $\alpha_{ij}^i \in [0, 2\pi)$ from the x -axis to the edge (i, j) in the body frame Σ_i , and the signed angles $\beta_{ij} = \beta_i - \beta_j$ from Σ_i to Σ_j , see Fig. 9 for illustrations. Similar settings for the measurement capability of agents can be found in [10, 44, 45] and the relative orientation can be obtained by an on board optical sensor [46]. We adopt a framework $(\mathcal{G}, p^*)$ to denote the target formation. The configuration set that has the same shape as the target formation is described as the set $\mathcal{E}(p^*)$.

Since ISAR ensures a unique shape of a framework, we consider the following assumption for the target formation.

Assumption 2: The target formation $(\mathcal{G}, p^*)$ is ISAR.

Note that $\mathcal{E}(p^*)$ is a 4-dimensional smooth manifold under Assumption 2. The signed angle-based formation control problem to be solved is stated below.

Problem 2: Given a set of signed angle constraints $\mathcal{S} = \{\alpha_{ijk}(p) = \alpha_{ijk}(p^*), (i, j, k) \in \mathcal{T}_{\mathcal{G}}\}$ induced by the framework $(\mathcal{G}, p^*)$, design a controller for each agent i based on only the signed angle measurements $\{\alpha_{ij}^i\}_{j \in \mathcal{N}_i}$ and $\{\beta_{ij}\}_{j \in \mathcal{N}_i}$ such that $\mathcal{E}(p^*)$ is asymptotically stable.

Under Assumption 2, the graph $\mathcal{G}$ must be connected. Therefore, any bearing in the framework $(\mathcal{G}, p^*)$ can be represented as the product of a rotation matrix determined by a linear combination of π and the elements in $\{\alpha_{ijk}^*\}_{(i,j,k) \in \mathcal{T}_{\mathcal{G}}}$ and a common bearing, which can be pre-specified, see Fig. 10 for an illustration. For simplicity, we utilize the symbol α_{ij}^* to denote the signed angle from the common edge $(1, 2)$ to the edge (i, j) in the desired framework $(\mathcal{G}, p^*)$, i.e., $\alpha_{ij}^* = s\pi + \sum_{(i,j,k) \in \mathcal{T}_{\mathcal{G}}} c_{ijk} \alpha_{ijk}^*$ for some $s, c_{ijk} \in \{0, \pm 1\}$. Therefore, the constraint sets $\{\alpha_{ij}^*\}_{(i,j) \in \mathcal{E}}$ and $\{\alpha_{ijk}^*\}_{(i,j,k) \in \mathcal{T}_{\mathcal{G}}}$ determine the same target formation $(\mathcal{G}, p^*)$ regardless of uniform rotations.

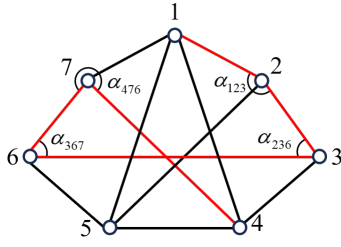


Fig. 10. The framework with an ISAR Laman subframework. Based on the common bearing $b_{12}(p^*)$, any other bearings can be denoted by the product of a rotation matrix determined by the marked signed angles and $b_{12}(p^*)$. For example, $b_{47}(p^*)$ can be depicted as $b_{47}(p^*) = \mathcal{R}_o(\alpha_{47}^*)b_{12}(p^*)$ where $\alpha_{47}^* = \pi + \alpha_{123} + \alpha_{236} + \alpha_{367} - \alpha_{476}$.

A. The Control Law and Stability Analysis

Inspired by the controllers in [10], we propose the following signed angle-based control law:

$$u_i^p(t) = - \sum_{j \in \mathcal{N}_i} \bar{R}(\alpha_{ij}^i) \eta(\beta_{ij}, \alpha_{ij}^*), \quad (21)$$

$$u_i^a(t) = - \sum_{j \in \mathcal{N}_i} \beta_{ij}, \quad (22)$$

where $\eta(\beta_{ij}, \alpha_{ij}^*) = \mathcal{R}_o(\frac{2\alpha_{ij}^* - \beta_{ij}}{2})[0, \cos \frac{\beta_{ij}}{2}]^\top$ and

$$\bar{R}(\alpha_{ij}^i) = \begin{bmatrix} 1 - \cos 2\alpha_{ij}^i & -\sin 2\alpha_{ij}^i \\ -\sin 2\alpha_{ij}^i & 1 + \cos 2\alpha_{ij}^i \end{bmatrix}.$$

Theorem 11 (Global Convergence): Given a group of agents with dynamics (19)–(20) moving in the plane and the target formation $(\mathcal{G}, p^*)$ satisfying Assumption 2. By implementing the controller (21)–(22), $p(t)$ converges to $\mathcal{E}(p^*)$ exponentially for arbitrary initial states $p(0)$ and $\beta(0)$.

Proof: See Appendix VIII-J. ■

Theorem 11 implies that the effectiveness of controller (21)–(22) only requires the sensing graph $\mathcal{G}$ of the formation to contain a Laman subgraph, which is more general than the required sensing graph⁴ structure outlined in the literature. For example, the sensing graph in Fig. 10 does not satisfy the condition in [11, 12], but satisfies Assumption 2.

Example 2: Consider a seven-agent formation, whose target formation shape is described in Fig. 10. The initial states $\{p_i(0)\}_{i \in \mathcal{V}}$ and $\{\beta_i(0)\}_{i \in \mathcal{V}}$ are assigned randomly. By implementing the controller (21)–(22), formation trajectories of all agents are shown in Fig. 11(a), and the evolution of signed angles errors is shown in Fig. 11(b). From the simulation results, we observe that the formation achieves the desired shape asymptotically, which is consistent with Theorem 11.

VII. CONCLUSION

In this paper, signed angle rigidity was reformulated by focusing on graphs. By establishing the complete equivalence between signed angle rigidity and bearing rigidity, we proved

⁴Although the authors in [12] utilized angularities to discuss rigidity, when being applied to formation control, there is always an underlying sensing graph describing the required sensing relationship among agents.

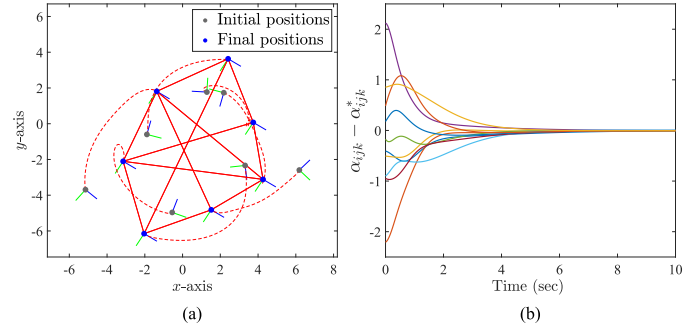


Fig. 11. (a) Trajectories of seven agents from randomly initial positions to final positions. (b) Evolution of signed angles errors.

that the shape of Laman graphs can be uniquely determined by signed angles for almost all configurations. In order to exclude redundant angle constraints in characterizing signed angle rigidity, an algorithm based on angle index graphs was developed to construct the minimal signed angle constraint set. Building on the developed signed angle rigidity theory, we further designed distributed network localization and formation stabilization algorithms, which are based on signed angle measurements only and achieve global exponential convergence.

However, the developed signed angle rigidity theory is limited to $\mathbb{R}^2$. Furthermore, the collision avoidance problem in formation control is not considered in this work, but is important in practice. Tackling these challenges will be a key focus of our future endeavors.

VIII. APPENDIX: PROOFS OF LEMMAS AND THEOREMS

A. Proof of Theorem 1

To prove Theorem 1, the following lemma will be used.

Lemma 8: In $\mathbb{R}^2$, a framework $(\mathcal{G}, p)$ is ISAR only if for any vertex $j \in \mathcal{V}$, there are at least two distinct neighbors $i, k \in \mathcal{N}_j$ such that $|b_{ji}^\top b_{jk}| \neq 1$.

Proof: We first show that each vertex has at least two different neighbors. Suppose there is a vertex $j \in \mathcal{V}$ with only one neighbor $i \in \mathcal{V}$. Let $q = [\mathbf{0}, \dots, \mathbf{0}, (b_{ji})^\top, \mathbf{0}, \dots, \mathbf{0}]^\top \in \mathbb{R}^{2n}$, where b_{ji} consists of the $(2j-1)$ -th to $2j$ -th components of q . According to Lemma 1, for any $\bar{\mathcal{T}}_G \subseteq \mathcal{T}_G$, one has

$$R_{\bar{\mathcal{T}}_G}^{\bar{\mathcal{T}}_G}(p)q = \bar{R}_{\bar{\mathcal{T}}_G}(p)\bar{H}q = 0. \quad (23)$$

Therefore, q is an infinitesimal signed angle motion but not trivial. Hence, a contradiction with the ISAR of $(\mathcal{G}, p)$ arises.

Similarly, if there is a vertex j satisfying $|\mathcal{N}_j| \geq 2$ and $|b_{ji}^\top b_{jk}| = 1$ for any pair of $i, k \in \mathcal{N}_j$, we can find a non-trivial infinitesimal signed angle motion $q = [\mathbf{0}, \dots, \mathbf{0}, (b_{ji})^\top, \mathbf{0}, \dots, \mathbf{0}]^\top \in \mathbb{R}^{2n}$ for some $i \in \mathcal{N}_j$, where b_{ji} consists of the $(2j-1)$ -th to $2j$ -th components of q . ■

We next prove Theorem 1.

Proof: From the rank condition of ISAR and IBR, it suffices to show that $\text{rank}(R_B(p)) = 2n - 3$ if and only if $\text{rank}(R_S^{\bar{\mathcal{T}}_G}(p)) = 2n - 4$.

According to [11, Theorem 5], we know that for a framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, $\text{rank}(R_B(p)) = 2n - 3$ if and only if $\text{rank}(R_A^{\mathcal{T}_G}(p)) = 2n - 4$, where $R_A^{\mathcal{T}_G}(p)$ is the rigidity matrix defined in [11] using unsigned angles (i.e., the cosine of the angles) as the entries of the rigidity function⁵. Therefore, it suffices to show that $\text{rank}(R_A^{\mathcal{T}_G}(p)) = 2n - 4$ if and only if $\text{rank}(R_S^{\mathcal{T}_G}(p)) = 2n - 4$.

From Lemma 8 and [11], we know that if $(\mathcal{G}, p)$ satisfies $\text{rank}(R_A^{\mathcal{T}_G}(p)) = 2n - 4$ or $\text{rank}(R_S^{\mathcal{T}_G}(p)) = 2n - 4$, then for any vertex $j \in \mathcal{V}$, there exist at least two distinct neighbors $i, k \in \mathcal{N}_j$ such that $|b_{ji}^\top b_{jk}| \neq 1$. As a result, when the framework satisfies $\text{rank}(R_A^{\mathcal{T}_G}(p)) = 2n - 4$ or $\text{rank}(R_S^{\mathcal{T}_G}(p)) = 2n - 4$, we can find an angle connected set $\bar{\mathcal{T}}_G \subseteq \mathcal{T}_G$ by [11, Algorithm 1], which satisfies $\sin \alpha_{ijk} \neq 0$ for $(i, j, k) \in \bar{\mathcal{T}}_G$ and $\text{rank}(R_A^{\bar{\mathcal{T}}_G}(p)) = \text{rank}(R_A^{\mathcal{T}_G}(p))$. Moreover, with virtue of Lemma 7 (will be proposed later with its proof independence of this theorem), one has $\text{rank}(R_S^{\bar{\mathcal{T}}_G}(p)) = \text{rank}(R_S^{\mathcal{T}_G}(p))$.

Necessity. We observe from the angle rigidity matrix in [11] and equation (6) that

$$R_S^{\mathcal{T}_G}(p) = \text{diag} \left(-\frac{1}{\sin \alpha_{ijk}} \right) R_A^{\mathcal{T}_G}(p), \quad (24)$$

where $\sin \alpha_{ijk} \neq 0$ for $(i, j, k) \in \bar{\mathcal{T}}_G$. Since $\text{rank}(R_A^{\bar{\mathcal{T}}_G}(p)) = \text{rank}(R_A^{\mathcal{T}_G}(p)) = 2n - 4$, one has

$$\text{rank}(R_S^{\bar{\mathcal{T}}_G}(p)) = \text{rank}(R_A^{\bar{\mathcal{T}}_G}(p)) = 2n - 4. \quad (25)$$

Together with the fact that $\text{rank}(R_S^{\mathcal{T}_G}(p)) = \text{rank}(R_S^{\bar{\mathcal{T}}_G}(p))$, we have $\text{rank}(R_S^{\mathcal{T}_G}(p)) = 2n - 4$.

Sufficiency. Similar to the discussion in the proof of necessity, it follows from $\text{rank}(R_S^{\mathcal{T}_G}(p)) = \text{rank}(R_S^{\bar{\mathcal{T}}_G}(p)) = 2n - 4$ and (24) that $\text{rank}(R_A^{\mathcal{T}_G}(p)) = 2n - 4$. ■

B. Proof of Lemma 3

Proof: Necessity. Note that for a signed angle α_{ijk} , if $\alpha_{ijk}(q) = \alpha_{ijk}(p)$ and $b_{ji}(q) = \mathcal{R}_o(\theta)b_{ji}(p)$ for some $\theta \in [0, 2\pi)$, one has $b_{jk}(q) = \mathcal{R}_o(\theta)b_{jk}(p)$. Consider any pair of vertices $i, j \in \mathcal{V}$, there exists an undirected path $\mathcal{P} = (\mathcal{V}_P, \mathcal{E}_P)$ between i and j due to the connectivity of $\mathcal{G}$. It follows from $S_G(q) = S_G(p)$ that $b_{l_{k-1}l_k}(q) = \mathcal{R}_o(\theta)b_{l_{k-1}l_k}(p)$ for any $(l_{k-1}, l_k) \in \mathcal{E}_P$ and some $\theta \in [0, 2\pi)$. Based on the connectivity of $\mathcal{G}$, we have $B_G(q) = [I_{|\mathcal{E}|} \otimes \mathcal{R}_o(\theta)] B_G(p)$.

Sufficiency. For any $(i, j), (j, k) \in \mathcal{E}$, it follows from $B_G(q) = [I_{|\mathcal{E}|} \otimes \mathcal{R}_o(\theta)] B_G(p)$ that

$$\begin{aligned} b_{jk}^\top(q)b_{ji}(q) &= b_{jk}^\top(p)\mathcal{R}_o^\top(\theta)\mathcal{R}_o(\theta)b_{ji}(p) \\ &= b_{jk}^\top(p)b_{ji}(p), \end{aligned} \quad (26)$$

$$\begin{aligned} b_{jk}^\top(q)\mathcal{R}_o(\frac{\pi}{2})b_{ji}(q) &= b_{jk}^\top(p)\mathcal{R}_o^\top(\theta)\mathcal{R}_o(\frac{\pi}{2})\mathcal{R}_o(\theta)b_{ji}(p) \\ &= b_{jk}^\top(p)\mathcal{R}_o(\frac{\pi}{2})b_{ji}(p). \end{aligned} \quad (27)$$

As a result, one has $S_G(q) = S_G(p)$ according to (5). ■

⁵In this paper, we use the triple (i, j, k) to index the unsigned angle between b_{ji} and b_{jk} , which is parallel with the definition in [11].

C. Proof of Theorem 2

Proof: Sufficiency. Suppose $(\mathcal{G}, p)$ is SAR. Then, there exists a neighborhood U_p of p such that for any $q \in S_G^{-1}(S_G(p)) \cap U_p$, it always holds $S_K(q) = S_K(p)$. It suffices to show that for this U_p , consider any $q \in B_G^{-1}(B_G(p)) \cap U_p$, it always holds $B_K(q) = B_K(p)$.

From $B_G(q) = B_G(p)$ and (5), we have $S_G(q) = S_G(p)$, which also implies $S_K(q) = S_K(p)$. Since $b_{ji}(p) = b_{ji}(q)$, $(j, i) \in \mathcal{E}$ and all signed angles of $\mathcal{K}(p)$ are determined, we can conclude that $B_K(q) = B_K(p)$.

Necessity. Suppose $(\mathcal{G}, p)$ is BR. Then, there exists a neighborhood U_p of p such that for any $q \in B_G^{-1}(B_G(p)) \cap U_p$, it always holds $B_K(q) = B_K(p)$. It suffices to show that for this U_p , consider any $q \in S_G^{-1}(S_G(p)) \cap U_p$, it always holds $S_K(q) = S_K(p)$. Note that the embedded graph of the BR framework must be connected, then it follows from Lemma 3 and $S_G(q) = S_G(p)$ that $B_G(q) = [I_{|\mathcal{E}|} \otimes \mathcal{R}_o(\theta)] B_G(p)$ for some $\theta \in [0, 2\pi)$. Since the framework is BR, it follows that $B_K(q) = [I_{|\mathcal{E}|} \otimes \mathcal{R}_o(\theta)] B_K(p)$. Thus, for any $i, j, k \in \mathcal{V}$,

$$\begin{aligned} b_{jk}^\top(q)b_{ji}(q) &= b_{jk}^\top(p)\mathcal{R}_o^\top(\theta)\mathcal{R}_o(\theta)b_{ji}(p) \\ &= b_{jk}^\top(p)b_{ji}(p), \end{aligned} \quad (28)$$

$$\begin{aligned} b_{jk}^\top(q)\mathcal{R}_o(\frac{\pi}{2})b_{ji}(q) &= b_{jk}^\top(p)\mathcal{R}_o^\top(\theta)\mathcal{R}_o(\frac{\pi}{2})\mathcal{R}_o(\theta)b_{ji}(p) \\ &= b_{jk}^\top(p)\mathcal{R}_o(\frac{\pi}{2})b_{ji}(p). \end{aligned} \quad (29)$$

According to (5), we further have $S_K(q) = S_K(p)$.

Similar to the proof of the equivalence between BR and SAR, we can derive the same result reflected on GBR and GSAR by extending U_p to $\mathbb{R}^{2n}$. ■

D. Proof of Theorem 3

Proof: (i) $\Leftrightarrow$ (iii). Suppose $(\mathcal{G}, p)$ is ISAR. From Lemma 4 and $\mathcal{E}(p) \subseteq \mathcal{E}_G(p)$, it suffices to show that for any $q \in \mathcal{E}_G(p)$, it always holds $q \in \mathcal{E}(p)$. Since $\mathcal{G}$ must be connected, it follows from Lemma 3 that $q \in \mathcal{E}_G(p)$ if only if $[I_n \otimes \mathcal{R}_o(\theta)]^{-1}q \in B_G^{-1}(B_G(p))$ for some $\theta \in [0, 2\pi)$. From [10, Theorem 6], for a given IBR framework $(\mathcal{G}, p)$, one has

$$B_G^{-1}(B_G(p)) = \{q \in \mathbb{R}^{2n} : q = cp + \mathbf{1}_n \otimes \xi, c \in \mathbb{R}^+, \xi \in \mathbb{R}^2\}. \quad (30)$$

Since ISAR is equivalent to IBR, then we have

$$[I_n \otimes \mathcal{R}_o(\theta)]^{-1}q = cp + \mathbf{1}_n \otimes \xi, \quad (31)$$

for some $c \in \mathbb{R}^+$ and $\xi \in \mathbb{R}^2$. As a result, $q \in \mathcal{E}(p)$.

Suppose the shape of $(\mathcal{G}, p)$ can be uniquely determined by signed angles up to uniform rotations, translations, and scalings. Then, p must be non-degenerate, and one has $\mathcal{E}(p) = \mathcal{E}_G(p)$ according to Lemma 4. Therefore, both $\mathcal{E}(p)$ and $\mathcal{E}_G(p)$ are 4-dimensional smooth manifolds. According to [47, Proposition 3.10], the tangent space of $\mathcal{E}_G(p)$ at p is 4-dimensional, which indicates that $\text{rank}(R_S^{\mathcal{T}_G}(p)) = 2n - 4$.

(ii) $\Leftrightarrow$ (iii). Suppose $(\mathcal{G}, p)$ is non-degenerate and GSAR. It suffices to show $\mathcal{E}(p) = \mathcal{E}_{\mathcal{G}}(p)$. Since $\mathcal{E}_{\mathcal{G}}(p) = \mathcal{E}_{\mathcal{K}}(p)$ and (i) $\Leftrightarrow$ (iii), we just need to show that there exists an ISAR spanning subframework of $(\mathcal{K}, p)$. From the non-degeneracy of $(\mathcal{K}, p)$, without loss of generality, we can select three non-collinear vertices from $(\mathcal{K}, p)$ and index them by 1, 2, 3. It can be observed from Lemma 2 that the framework $(\mathcal{G}', p')$ embedded by the triangle $\mathcal{G}'$ composed of 1, 2, 3 is ISAR, where $p' \in \mathbb{R}^6$ is the configuration of 1, 2, 3 in p . Since for any $4 \leq j \leq n$, there always exist $i, k \in \{1, 2, 3\}$ such that b_{ji} and b_{jk} are not collinear. Combining [11, Theorem 3], Theorem 1, and [17, Lemma 7], we conclude that the framework embedded by the graph constructed by adding vertex j and edges (j, i) , (j, k) to $\mathcal{G}'$ for $j = 4, \dots, n$ iteratively is ISAR.

Suppose $\mathcal{E}(p) = \mathcal{E}_{\mathcal{G}}(p)$. Then, p must be non-degenerate, and hence, it suffices to show that for any $q \in \mathcal{E}_{\mathcal{G}}(p)$, it always holds $q \in \mathcal{E}_{\mathcal{K}}(p)$. Since $\mathcal{E}(p) = \mathcal{E}_{\mathcal{G}}(p)$ and $q \in \mathcal{E}_{\mathcal{G}}(p)$, we have

$$q = c[I_n \otimes \mathcal{R}_o(\theta)]p + \mathbf{1}_n \otimes \xi \quad (32)$$

for some $\theta \in [0, 2\pi)$, $c \in \mathbb{R}^+$, and $\xi \in \mathbb{R}^2$. Therefore, one has $B_{\mathcal{K}}(q) = [I_{|\mathcal{E}|} \otimes \mathcal{R}_o(\theta)] B_{\mathcal{K}}(p)$ and $S_{\mathcal{K}}(q) = S_{\mathcal{K}}(p)$. ■

E. Proof of Theorem 7

Proof: According to Theorem 3 and Lemma 5, for an ISAR framework $(\mathcal{G}, p)$ in $\mathbb{R}^2$, $\bar{\mathcal{T}}_{\mathcal{G}} \subseteq \mathcal{T}_{\mathcal{G}}$ is a GAIS if and only if $\mathcal{E}_{\bar{\mathcal{T}}_{\mathcal{G}}}(p) = \mathcal{E}(p)$. Since p is non-degenerate, $\mathcal{E}_{\bar{\mathcal{T}}_{\mathcal{G}}}(p)$ is a 4-dimensional smooth manifold. According to [47, Proposition 3.10], the tangent space of $\mathcal{E}_{\bar{\mathcal{T}}_{\mathcal{G}}}(p)$ at p is 4-dimensional. Hence, $\text{rank}(R_S^{\bar{\mathcal{T}}_{\mathcal{G}}}(p)) = 2n - 4$ and $|\bar{\mathcal{T}}_{\mathcal{G}}| \geq 2n - 4$.

The sufficiency follows from Lemma 6 and the fact that $|\bar{\mathcal{T}}_{\mathcal{L}_n}|$ is lower bounded by $2n - 4$. To prove necessity, we denote the angle index graph of $\bar{\mathcal{T}}_{\mathcal{L}_n}$ and $\mathcal{L}_n = (\mathcal{V}_n, \mathcal{E}_n)$ by $(\mathcal{V}_A, \mathcal{E}_A)$ and show the following two cases contradict the fact that $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is minimal to satisfy $\mathcal{E}_{\bar{\mathcal{T}}_{\mathcal{L}_n}}(p) = \mathcal{E}_{\mathcal{L}_n}(p)$: (i) $|\bar{\mathcal{T}}_{\mathcal{L}_n}| > 2n - 4$ and $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is angle connected; (ii) $|\bar{\mathcal{T}}_{\mathcal{L}_n}| \geq 2n - 4$ and $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is not angle connected.

Case 1. Since $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is angle connected, we can use the breadth first search method to find a spanning tree within $(\mathcal{V}_A, \mathcal{E}_A)$, it follows from Lemma 6 that the AIS $\bar{\mathcal{T}}_{\mathcal{L}_n}^\dagger$ corresponding to the spanning tree satisfies $\mathcal{E}_{\bar{\mathcal{T}}_{\mathcal{L}_n}^\dagger}(p) = \mathcal{E}_{\mathcal{L}_n}(p)$ and $|\bar{\mathcal{T}}_{\mathcal{L}_n}^\dagger| = |\mathcal{E}_n| - 1$. Since the embedded graph is a Laman graph, one has $|\mathcal{E}_n| = 2n - 3$, and consequently,

$$|\bar{\mathcal{T}}_{\mathcal{L}_n}^\dagger| = |\mathcal{E}_n| - 1 = 2n - 4 < |\bar{\mathcal{T}}_{\mathcal{L}_n}|. \quad (33)$$

Hence, $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is not minimal for $\mathcal{E}_{\bar{\mathcal{T}}_{\mathcal{L}_n}}(p) = \mathcal{E}_{\mathcal{L}_n}(p)$.

Case 2. Without loss of generality, $(\mathcal{V}_A, \mathcal{E}_A)$ has only two connected components $(\mathcal{V}_A^1, \mathcal{E}_A^1)$ and $(\mathcal{V}_A^2, \mathcal{E}_A^2)$, where $\mathcal{V}_A^1 \cup \mathcal{V}_A^2 = \mathcal{V}_A$ and $\mathcal{E}_A^1 \cup \mathcal{E}_A^2 = \mathcal{E}_A$. It follows from $|\mathcal{E}_A| = |\bar{\mathcal{T}}_{\mathcal{L}_n}| \geq 2n - 4$ and $|\mathcal{V}_A| = |\mathcal{E}_n| = 2n - 3$ that $|\mathcal{E}_A| > |\mathcal{V}_A| - 2$. Since the minimum number of edges of a 2-component graph with n vertices is $n - 2$, there exists an undirected cycle in $(\mathcal{V}_A, \mathcal{E}_A)$. Suppose that the undirected cycle is in $(\mathcal{V}_A^1, \mathcal{E}_A^1)$. Denote it by $\mathcal{C} = (\mathcal{V}_C, \mathcal{E}_C)$, where

$\mathcal{V}_C \subseteq \mathcal{V}_A^1$ and $\mathcal{E}_C \subseteq \mathcal{E}_A^1$. Since constraining signed angles α_{ijk} corresponding to $(a_{ji}, a_{jk}) \in \mathcal{E}_C$ only requires $|\mathcal{V}_C| - 1 < |\mathcal{E}_C|$ signed angle constraints, the signed angle constraints given by $\bar{\mathcal{T}}_{\mathcal{L}_n}$ are not minimal, which causes a contradiction.

As a result, $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is minimal to satisfy $\mathcal{E}_{\bar{\mathcal{T}}_{\mathcal{L}_n}}(p) = \mathcal{E}_{\mathcal{L}_n}(p)$ only if $|\bar{\mathcal{T}}_{\mathcal{L}_n}| = 2n - 4$ and $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is angle connected. ■

F. Proof of Lemma 7

Proof: According to Definition 7, it is sufficient to show $\text{rank}(R_S^{\bar{\mathcal{T}}_{\mathcal{G}}}(p)) = \text{rank}(R_S^{\mathcal{T}_{\mathcal{G}}}(p))$. Since $\bar{\mathcal{T}}_{\mathcal{G}}$ is angle connected, for any two edges $(i, j), (l, m) \in \mathcal{E}$, one has $b_{lm} = \mathcal{R}_o(\alpha_{lm}^*)b_{ij}$ (see Fig. 10 as an example), where

$$\alpha_{lm}^* = s\pi + \sum_{(i,j,k) \in \bar{\mathcal{T}}_{\mathcal{G}}} c_{ijk}\alpha_{ijk} \quad (34)$$

for some $c_{ijk}, s \in \{\pm 1, 0\}$. Therefore, the signed angle $\alpha \in [0, 2\pi)$ between any pair of bearings in $(\mathcal{G}, p)$ satisfies

$$s_1\pi + \alpha = \begin{cases} \alpha^* & \text{if } \sin \alpha^* \geq 0, \\ 2\pi - \alpha^* & \text{otherwise,} \end{cases} \quad (35)$$

where $\alpha^* = s_2\pi + \sum_{(i,j,k) \in \bar{\mathcal{T}}_{\mathcal{G}}} c_{ijk}\alpha_{ijk}$ for some $s_2, c_{ijk} \in \{0, \pm 1\}$ and $s_1 \in \mathbb{N}$. Differentiating both sides of (35) w.r.t. time leads to $\dot{\alpha} = s_3 \sum_{(i,j,k) \in \bar{\mathcal{T}}_{\mathcal{G}}} c_{ijk}\dot{\alpha}_{ijk}$ for some $s_3 \in \{\pm 1\}$, which means that any row in $R_S^{\bar{\mathcal{T}}_{\mathcal{G}}}(p)$ can be represented as a linear combination of the rows indexed by $\bar{\mathcal{T}}_{\mathcal{G}}$. As a result, $\text{rank}(R_S^{\bar{\mathcal{T}}_{\mathcal{G}}}(p)) = \text{rank}(R_S^{\mathcal{T}_{\mathcal{G}}}(p))$. ■

G. Proof of Theorem 8

Proof: Combining with Theorem 7, it is sufficient to show that for an ISAR Laman framework $(\mathcal{L}_n, p)$ in $\mathbb{R}^2$, $\bar{\mathcal{T}}_{\mathcal{L}_n} \subseteq \mathcal{T}_{\mathcal{L}_n}$ is a minimal RAIS if and only if $|\bar{\mathcal{T}}_{\mathcal{L}_n}| = 2n - 4$ and $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is angle connected. We first note that for an ISAR framework $(\mathcal{G}, p)$, $\bar{\mathcal{T}}_{\mathcal{G}} \subseteq \mathcal{T}_{\mathcal{G}}$ is a RAIS if and only if $\text{rank}(R_S^{\bar{\mathcal{T}}_{\mathcal{G}}}(p)) = \text{rank}(R_S^{\mathcal{T}_{\mathcal{G}}}(p)) = 2n - 4$, which implies that $|\bar{\mathcal{T}}_{\mathcal{G}}| \geq 2n - 4$.

The sufficiency follows from Lemma 7 and $|\bar{\mathcal{T}}_{\mathcal{L}_n}| \geq 2n - 4$. As for necessity, by employing Lemma 7 and the techniques used in the proof of Theorem 7, it can be verified that the following two cases contradict the fact that $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is minimal to satisfy $\text{rank}(R_S^{\bar{\mathcal{T}}_{\mathcal{L}_n}}(p)) = \text{rank}(R_S^{\mathcal{T}_{\mathcal{L}_n}}(p))$: (i) $|\bar{\mathcal{T}}_{\mathcal{L}_n}| > 2n - 4$ and $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is angle connected; (ii) $|\bar{\mathcal{T}}_{\mathcal{L}_n}| \geq 2n - 4$ and $\bar{\mathcal{T}}_{\mathcal{L}_n}$ is not angle connected. ■

H. Proof of Theorem 9

Proof: Sufficiency. Let $q = [q_1^\top, \dots, q_n^\top]^\top \in \mathbb{R}^{2n}$ be a solution to (15). Then, we have $q \in \mathcal{E}_{\mathcal{G}}(p)$. It suffices to show $q = p$. Since $(\mathcal{G}, p)$ is ISAR, it follows from Theorem 3 that

$$q = c[I_n \otimes \mathcal{R}_o(\theta)]p + \mathbf{1}_n \otimes \xi, \quad (36)$$

for some $c \in \mathbb{R}^+$, $\theta \in [0, 2\pi)$, and $\xi \in \mathbb{R}^2$. Without loss of generality, let 1, 2 be the anchors since $|\mathcal{A}| \geq 2$. It follows from $q_i = p_i, i \in \mathcal{A}$ that

$$p_1 - p_2 = q_1 - q_2 = c\mathcal{R}_o(\theta)(p_1 - p_2), \quad (37)$$

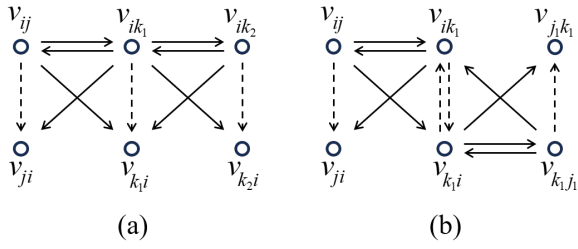


Fig. 12. Illustrations of partial topology relationships of $\hat{\mathcal{G}}$. (a) There exist $(a_{ij}, a_{ik_1}), (a_{ik_1}, a_{ik_2}) \in \mathcal{E}_A$. (b) There exist $(a_{ij}, a_{ik_1}), (a_{ik_1}, a_{k_1j_1}) \in \mathcal{E}_A$. The solid arrow represents the directed edge in $\hat{\mathcal{G}}$ and the dashed arrow represents the existence of a directed path from the starting node to the terminal node.

which implies $c\mathcal{R}_o(\theta) = I_2$. Together with the fact that $q_1 = c\mathcal{R}_o(\theta)p_1 + \xi = p_1$, we have $\xi = [0, 0]^\top$ and $q = p$. Hence, $(\mathcal{G}, p, \mathcal{A})$ is signed-angle-based localizable.

Necessity. It suffices to show that there exists a solution $q \in \mathbb{R}^{2n}$ of (15) such that $q \neq p$ when $|\mathcal{A}| \leq 1$. Without loss of generality, suppose $\mathcal{A} = \{1\}$. Let $q = [I_n \otimes \mathcal{R}_o(\frac{\pi}{2})]p + \mathbf{1}_n \otimes \eta$, where $\eta = [I_2 - \mathcal{R}_o(\frac{\pi}{2})]p_1$. One has $q_1 = p_1$, $q \neq p$, and $q \in \mathcal{E}_g(p)$. On the other hand, when $\mathcal{A} = \emptyset$ and $q = p + \mathbf{1}_n \otimes \xi$ with any vector $\xi \in \mathbb{R}^2$ is a solution of (15). ■

I. Proof of Theorem 10

To show convergence, similar to [48, pp. 110], for the network $(\mathcal{G}, p, \mathcal{A})$ and the AIS $\bar{\mathcal{T}}_{\mathcal{G}}$, we define $\hat{\mathcal{G}} \triangleq (\hat{\mathcal{V}}, \hat{\mathcal{E}})$ as the bearing graph under (18), which is a directed graph describing the pairwise relationship between bearings in the angle errors feedback according to (18). Here, $v_{ij} \in \hat{\mathcal{V}}$ indexes the estimate $\hat{b}_{ij}(t)$ for $i \in \mathcal{V}$ and $j \in \mathcal{N}_G^i$ and is computed by

$$v_{ij} = \begin{cases} 2\mathcal{H}(i, j, |\mathcal{V}|) & \text{if } i < j, \\ 2\mathcal{H}(i, j, |\mathcal{V}|) + 1 & \text{otherwise,} \end{cases} \quad (38)$$

a directed edge (v_{ik}, v_{ij}) from v_{ik} to v_{ij} exists in $\hat{\mathcal{E}}$ if $\hat{b}_{ik}(t)$ contributes to the update of $\hat{b}_{ij}(t)$, and $(v_{jk}, v_{ij}) \in \hat{\mathcal{E}}$ if $\hat{b}_{jk}(t)$ is used in the update of $\hat{b}_{ij}(t)$. Let $\bar{\mathcal{A}}$ be the leader set of the angle index graph $(\mathcal{V}_A(\mathcal{G}), \mathcal{E}_A(\bar{\mathcal{T}}_{\mathcal{G}}))$, where $a_{ij} \in \mathcal{V}_A(\mathcal{G})$ belongs to $\bar{\mathcal{A}}$ if $i, j \in \mathcal{A}$. Let $\hat{\mathcal{A}} = \{v_{ij}, v_{ji} : a_{ij} \in \bar{\mathcal{A}}\}$ be the leader set of $\hat{\mathcal{G}}$. Under Assumption 1, we know that $\hat{\mathcal{A}} \neq \emptyset$.

Similarly to the undirected path, given the directed graph $\hat{\mathcal{G}}$, a directed path from $l_1 \in \hat{\mathcal{V}}$ to $l_{n_P} \in \hat{\mathcal{V}}$ is a sequence of directed edges of the form $(l_1, l_2), \dots, (l_{n_P-1}, l_{n_P})$.

Next, we provide a useful lemma with regard to $\hat{\mathcal{G}}$.

Lemma 9: Given the network $(\mathcal{G}, p, \mathcal{A})$ satisfying Assumption 1 and the angle connected AIS $\bar{\mathcal{T}}_{\mathcal{G}}$ induced by Algorithm 1, for any pair of elements $a_{ij}, a_{lk} \in \mathcal{V}_A(\mathcal{G}) \setminus \bar{\mathcal{A}}$ and the bearing graph $\hat{\mathcal{G}}$ under (18), there exists a directed path from v_{ij} to v_{lk} or from v_{ji} to v_{lk} in $\hat{\mathcal{G}}$ if there exists an undirected path $\mathcal{P}_A = (\mathcal{V}_P^A, \mathcal{E}_P^A)$ satisfying $\mathcal{V}_P^A \cap \bar{\mathcal{A}} = \emptyset$ between a_{ij} and a_{lk} in the angle index graph $\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{G}}) = (\mathcal{V}_A(\mathcal{G}), \mathcal{E}_A(\bar{\mathcal{T}}_{\mathcal{G}}))$.

Proof: For simplicity of notation, denote $\mathcal{G}_A(\bar{\mathcal{T}}_{\mathcal{G}}) = (\mathcal{V}_A(\mathcal{G}), \mathcal{E}_A(\bar{\mathcal{T}}_{\mathcal{G}}))$ by $\mathcal{G}_A = (\mathcal{V}_A, \mathcal{E}_A)$. We first note that for any pair of $a_{ij}, a_{ik_1} \in \mathcal{V}_A \setminus \bar{\mathcal{A}}$ with $(a_{ij}, a_{ik_1}) \in \mathcal{E}_A$, it

follows from (18) and the definition of $\hat{\mathcal{E}}$ that the directed edges $(v_{ij}, v_{ik_1}), (v_{ik_1}, v_{ij}), (v_{ik_1}, v_{ji}),$ and (v_{ij}, v_{k_1i}) belong to $\hat{\mathcal{E}}$ (see Fig. 12 for examples). It also implies that v_{ij} and v_{ik_1} are mutually reachable in $\hat{\mathcal{G}}$ and there are two directed paths from v_{ij} to v_{ji} and from v_{ik_1} to v_{k_1i} , respectively.

Given any pair of $a_{ij}, a_{lk} \in \mathcal{V}_A \setminus \bar{\mathcal{A}}$. If a_{ij} is adjacent to a_{lk} in $\mathcal{G}_A$, then the conclusion holds, immediately. If a_{ij} is not adjacent to a_{lk} in $\mathcal{G}_A$ and there exists an undirected path $\mathcal{P}_A = (\mathcal{V}_P^A, \mathcal{E}_P^A)$ between a_{ij} and a_{lk} , where $\mathcal{V}_P^A \subseteq \mathcal{V}_A \setminus \bar{\mathcal{A}}$, $\mathcal{E}_P^A \subseteq \mathcal{E}_A$, and $|\mathcal{V}_P^A| \geq 3$. Without loss of generality, suppose that $(a_{ij}, a_{ik_1}) \in \mathcal{E}_P^A$. The rest of the proof is to show that if a_{gh} is a neighbor of a_{ik_1} (other than a_{ij}) in $\mathcal{P}_A$, there exists two directed paths from v_{ij} to v_{gh} and to v_{hg} , respectively. Then, the statement can be proved by induction.

Since the number of neighbors of a_{ik_1} in $\mathcal{P}_A$ is two, one of the following two cases holds:

Case 1: $\exists(a_{ik_1}, a_{ik_2}) \in \mathcal{E}_P^A$. Since $(a_{ij}, a_{ik_1}), (a_{ik_1}, a_{ik_2}) \in \mathcal{E}_A$, it follows that for any pair of vertices in $\{v_{ij}, v_{ik_1}, v_{ik_2}\}$, they are mutually reachable in $\hat{\mathcal{G}}$ (see Fig. 12(a)), and there are two directed paths from v_{ij} to v_{ik_2} and to v_{k_2i} , respectively.

Case 2: $\exists(a_{ik_1}, a_{k_1j_1}) \in \mathcal{E}_P^A$. It follows from $(a_{ij}, a_{ik_1}), (a_{ik_1}, a_{k_1j_1}) \in \mathcal{E}_A$ that for any pair of vertices in $\{v_{ik_1}, v_{k_1i}, v_{ij}, v_{k_1j_1}\}$, they are mutually reachable in $\hat{\mathcal{G}}$ (see Fig. 12(b)), and there are two directed paths from v_{ij} to $v_{k_1j_1}$ and to $v_{j_1k_1}$, respectively. ■

To prove Theorem 10, the following lemma will be used.

Lemma 10: Given the network $(\mathcal{G}, p, \mathcal{A})$ satisfying Assumption 1 and the angle connected AIS $\bar{\mathcal{T}}_{\mathcal{G}}$ induced by Algorithm 1, there exists a path from some vertex in $\hat{\mathcal{A}}$ to v_{lk} for any vertex $v_{lk} \in \hat{\mathcal{V}} \setminus \hat{\mathcal{A}}$ under (18).

Proof: Let the angle index graph of $\mathcal{G}$ and $\bar{\mathcal{T}}_{\mathcal{G}}$ be $\mathcal{G}_A = (\mathcal{V}_A, \mathcal{E}_A)$. The rest of the proof is to show that under the cases (i) $|\bar{\mathcal{A}}| = 1$ and (ii) $|\bar{\mathcal{A}}| \geq 2$, the statement holds.

When $|\bar{\mathcal{A}}| = 1$, without loss of generality, let $\bar{\mathcal{A}} = \{a_{12}\}$, and suppose $(a_{12}, a_{1k}) \in \mathcal{E}_A$. Since $\mathcal{G}_A$ is a tree, we consider the following three cases:

Case 1: a_{12} is a leaf vertex and $\exists(a_{1k}, a_{1k_2}) \in \mathcal{E}_A$. Since $(v_{12}, v_{1k}) \in \hat{\mathcal{E}}$, it suffices to show that for any $v_{ij} \in \hat{\mathcal{V}} \setminus \hat{\mathcal{A}}$, there exists a directed path from v_{1k} to v_{ij} . Recall that $a_{1k}, a_{1k_2} \notin \bar{\mathcal{A}}$ and the graph $\bar{\mathcal{G}}_A = (\bar{\mathcal{V}}_A, \bar{\mathcal{E}}_A)$ formed by removing a_{12} from $\mathcal{G}_A$ is still a connected graph. Hence, there exists an undirected path $\mathcal{P}_A = (\mathcal{V}_P^A, \mathcal{E}_P^A)$ between a_{1k} and a_{ij} in the angle index graph $\mathcal{G}_A$, where $\mathcal{V}_P^A \subseteq \mathcal{V}_A \setminus \bar{\mathcal{A}}$ and $\mathcal{E}_P^A \subseteq \mathcal{E}_A$. It follows from Lemma 9 that there is a directed path from v_{1k} to v_{ij} or from v_{k_1} to v_{ij} in $\hat{\mathcal{G}}$. Since $(a_{1k}, a_{1k_2}) \in \mathcal{E}_A$ and $a_{1k}, a_{1k_2} \notin \bar{\mathcal{A}}$, there is a directed path from v_{1k} to v_{ij} and a directed path from v_{12} to v_{ij} .

Case 2: a_{12} is a leaf vertex and $\exists(a_{1k}, a_{k_1j_1}) \in \mathcal{E}_A$. Using techniques similar to Case 1, it can be verified that for any $v_{ij} \in \hat{\mathcal{V}} \setminus \hat{\mathcal{A}}$, there is a directed path from v_{12} to v_{ij} .

Case 3: a_{12} is a branch vertex. Let $\mathcal{N}_{a_{12}} = \{a_{12}^1, \dots, a_{12}^m\}$ be the neighbor set of a_{12} in $\mathcal{G}_A$. Since a_{12} is a branch vertex, $|\mathcal{N}_{a_{12}}| = m \geq 2$ and we can find m subtrees $\bar{\mathcal{G}}_A^i = (\bar{\mathcal{V}}_A^i, \bar{\mathcal{E}}_A^i)$ (for $i = 1, \dots, m$) from the tree $\mathcal{G}_A$, where $\bar{\mathcal{V}}_A^i = \mathcal{V}_A^i \cup \{a_{12}\}$,

$\mathcal{V}_A^i$ is the set of the vertices $a_{uv} \in \mathcal{V}_A$ that have a path from a_{12} to a_{uv} passing through a_{12}^i , and $\bar{\mathcal{E}}_A^i = \{(a_{uv}, a_{ef}) \in \mathcal{E}_A : a_{uv}, a_{ef} \in \bar{\mathcal{V}}_A^i\}$. Note that a_{12} is a leaf vertex in each subtree $\bar{\mathcal{G}}_A^i$. Therefore, it follows from Case 1–2 that for any $a_{jk} \in \bar{\mathcal{V}}_A^i \setminus \{a_{12}\}$, there exist two directed paths from v_{12} to v_{jk} and to v_{kj} , respectively.

When $|\bar{\mathcal{A}}| \geq 2$, one can form the graph $\bar{\mathcal{G}}_A = (\bar{\mathcal{V}}_A, \bar{\mathcal{E}}_A)$ by removing the vertices $a_{ij} \in \bar{\mathcal{A}}$ from $\mathcal{G}_A$. Let m be the number of connected components in $\bar{\mathcal{G}}_A$, and denote each connected component by $\bar{\mathcal{G}}_A^i = (\bar{\mathcal{V}}_A^i, \bar{\mathcal{E}}_A^i)$ (for $i = 1, \dots, m$). Since each component $\bar{\mathcal{G}}_A^i$ contains a vertex $a_{lk} \in \bar{\mathcal{V}}_A^i$ that is adjacent to some vertex $a_{uv} \in \bar{\mathcal{A}}$ in $\mathcal{G}_A$, we can further make $\bar{\mathcal{V}}_A^i = \bar{\mathcal{V}}_A^i \cup \{a_{uv}\}$ and $\bar{\mathcal{E}}_A^i = \bar{\mathcal{E}}_A^i \cup \{(a_{uv}, a_{lk})\}$. Then $\bar{\mathcal{G}}_A^i$ becomes a tree after this augmentation and has only one vertex $a_{uv} \in \bar{\mathcal{V}}_A^i$ belonging to $\bar{\mathcal{A}}$. The analysis of the case $|\bar{\mathcal{A}}| = 1$ shows that for any vertex $a_{lk} \in \bar{\mathcal{V}}_A^i \setminus \bar{\mathcal{A}}$, there exist two directed path from some vertex in $\bar{\mathcal{A}}$ to v_{lk} and to v_{kl} , respectively. ■

We next prove Theorem 10.

Proof: Note that according to Theorem 1 and [16, Corollary 3], Assumption 1 implies that $(\mathcal{G}, p, \mathcal{A})$ is bearing localizable [16]. Therefore, we first prove that $\hat{b}_{ij}(t)$ exponentially converges to the global bearing b_{ij} for all $j \in \mathcal{N}_i$. Then, the proof of the convergence of $\hat{p}_i(t)$ to p_i for all $i \in \mathcal{F}$ is similar to that in [43, Theorem 2].

Let $B_{ij} \triangleq [b_{ij}, b_{ij}^\perp]$. Given any $\alpha_{ijk} \in [0, 2\pi)$, it follows from $\mathcal{R}_o(\alpha_{jik})b_{ij} = b_{ik}$ and $\mathcal{R}_o(\alpha_{jik})b_{ij}^\perp = b_{ik}^\perp$ that $\mathcal{R}_o(\alpha_{jik})B_{ij} = B_{ik}$. Multiplying $B_{ij}^\top$ on both sides of (18),

$$\begin{aligned} B_{ij}^\top \dot{\hat{b}}_{ij} = & - \left\{ \sum_{(j,i,k_1) \in \bar{\mathcal{T}}_{\mathcal{G}}} [B_{ij}^\top \hat{b}_{ij} - B_{ik_1}^\top \hat{b}_{ik_1}] \right. \\ & + \sum_{(k_2,i,j) \in \bar{\mathcal{T}}_{\mathcal{G}}} [B_{ij}^\top \hat{b}_{ij} - B_{ik_2}^\top \hat{b}_{ik_2}] \\ & + \sum_{(i,j,k_3) \in \bar{\mathcal{T}}_{\mathcal{G}}} [B_{ij}^\top \hat{b}_{ij} - B_{jk_3}^\top \hat{b}_{jk_3}] \\ & \left. + \sum_{(k_4,j,i) \in \bar{\mathcal{T}}_{\mathcal{G}}} [B_{ij}^\top \hat{b}_{ij} - B_{jk_4}^\top \hat{b}_{jk_4}] \right\}, \quad (39) \end{aligned}$$

where we use the fact that $B_{ij} = -B_{ji}$. Let $x_{ij} \triangleq B_{ij}^\top \hat{b}_{ij}$ for $j \in \mathcal{N}_i$ and $\mathcal{N}_{v_{ij}}$ be the neighbor set of the vertex v_{ij} in the bearing graph $\hat{\mathcal{G}}$ under (18). Then, (39) can be rewritten as

$$\dot{x}_{ij} = - \left[\sum_{v_{ik} \in \mathcal{N}_{v_{ij}}} (x_{ij} - x_{ik}) + \sum_{v_{jk} \in \mathcal{N}_{v_{ij}}} (x_{ij} - x_{jk}) \right]. \quad (40)$$

Since for each vertex $v_{ij} \in \hat{\mathcal{V}} \setminus \hat{\mathcal{A}}$, there exists a directed path from some vertex in $\hat{\mathcal{A}}$ to v_{ij} by Lemma 10, and $x_{jk} = B_{jk}^\top \hat{b}_{jk} = B_{jk}^\top b_{jk} = [1, 0]^\top$ for $v_{jk} \in \hat{\mathcal{A}}$ and all $t \geq 0$, x_{ij} converges to $[1, 0]^\top$ for all $j \in \mathcal{N}_i$ according to the result in [48, Theorem 5.1]. It implies that the system converges to

$$\begin{cases} \hat{b}_{ij}^\top b_{ij} = \|\hat{b}_{ij}\| \cos \theta = 1, \\ \hat{b}_{ij}^\top b_{ij}^\perp = \|\hat{b}_{ij}\| \sin \theta = 0, \end{cases} \quad (41)$$

where θ is the signed angle from $\hat{b}_{ij}$ to b_{ij} . It follows from (41) that $\|\hat{b}_{ij}\| \mathcal{R}_o(\theta)[1, 0]^\top = [1, 0]^\top$. Therefore, one has $\|\hat{b}_{ij}\| = 1$ and $\theta = 0$. As a result, $\hat{b}_{ij} \rightarrow b_{ij}$ as $t \rightarrow \infty$ for all $j \in \mathcal{N}_i$. Since the system (40) is linear, the convergence is exponentially fast [49, Corollary 4.3]. ■

J. Proof of Theorem 11

Proof: Since $\beta_{ij} = \beta_i - \beta_j$, substituting (22) into (20) leads to,

$$\dot{\beta}_i = \sum_{j \in \mathcal{N}_i} (\beta_j - \beta_i). \quad (42)$$

It can be observed that (42) is the same as the consensus algorithm in [48, Lemma 1.3] under a connected undirected graph $\mathcal{G}$. Therefore, $\beta_{ij}, (i, j) \in \mathcal{E}$ converges to zero exponentially fast for any initial points $\beta_i(0) \in [0, 2\pi), i \in \mathcal{V}$. That is, the body frames of agents will achieve coordinate alignment exponentially. Using similar techniques in the proof of [10, Theorem 15], it can be concluded that the controller (21) will drive $b_{ij}(p), (i, j) \in \mathcal{E}$ to either $\mathcal{R}_o(\theta)b_{ij}(p^*)$ (for almost all initial values) or $-\mathcal{R}_o(\theta)b_{ij}(p^*)$ (for the remaining case) exponentially fast, where $\theta \in [0, 2\pi)$. It follows that $\alpha_{ijk}(p) \rightarrow \alpha_{ijk}(p^*), (i, j, k) \in \mathcal{T}_{\mathcal{G}}$ as $t \rightarrow \infty$ from any initial values. According to Theorem 3, the position $p(t)$ of the formation converges to the manifold $\mathcal{E}(p^*)$ globally. ■

IX. ACKNOWLEDGMENTS

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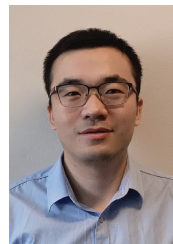
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